Mapping the Nutrition–Urological Cancer Risk Landscape: Large Language Model–Assisted Meta-analyses of Bladder, Kidney, and Prostate Cancers

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Abstract

Manual meta-analyses of diet and cancer are slow to produce and quick to age. We developed metaUro, a monthly updated web platform that uses large language models to synthesize evidence on over 400 nutritional exposures and the incidence of bladder, kidney and renal pelvis, and prostate cancers, providing on-demand summaries across exposures. For each exposure–cancer pair, PubMed records were screened and extracted by two independent language-model pipelines, discrepancies were adjudicated by team members, and estimates were pooled using random-effects meta-analysis. Benchmarking against two published meta-analyses of prospective studies, metaUro retrieved all 20 articles (12 and 8, respectively). Of these, seven (4 and 3, respectively) could not be pooled because essential quantitative data were inaccessible behind journal paywalls; one was missed during exposure-specific screening because the exposure was absent from its title and abstract; and two were excluded because metaUro adhered more closely to the prespecified eligibility criteria. Conversely, metaUro identified five additional eligible articles (3 and 2, respectively) not included in the benchmark meta-analyses. Across exposures, red meat was positively and selenium inversely associated with all three cancers, whereas alcohol was positively associated with prostate cancer but inversely with kidney and bladder cancers.

1 Introduction

For 2026, the American Cancer Society projects 333,830 new cases of prostate cancer and 80,450 new cases of kidney and renal pelvis cancer in the United States [1]. Alongside established non-nutritional risk factors, such as smoking [2, 3], nutritional factors have attracted increasing attention as potential targets for urological cancer prevention [4], underscoring the need for further research.

Systematic reviews and meta-analyses offer a structured path from individual nutritional studies to a summary estimate [5], but manually conducted analyses are slow to produce and quick to age. In one registry-based analysis, completing and publishing a review took 67 weeks on average [6]; in a cohort of high-quality reviews, 23% showed signs that an update was needed within 2 years [7]. Overlapping reviews of the same question are also common [8].

Large language models (LLMs) offer a promising approach to addressing the rapidly growing volume of scientific literature and the limited capacity of manual curation, which has become a major bottleneck in conducting timely systematic reviews. Recent studies [9, 10, 11, 12, 13, 14, 15, 16, 17] have explored the potential of LLM-assisted systematic reviews and evaluated the risks associated with this automation. However, these approaches primarily support one user-specified review at a time. As summarized in Table 1, they may also require users to configure software, establish paid LLM API accounts, provide computing resources, or purchase commercial access. In contrast, the proposed platform, metaUro, (accessible through `https://metauro.up.railway.app/`) requires no API keys, software installation, or user-provided computing resources, making it readily and freely accessible to biomedical researchers. By synthesizing nutritional evidence across 446 exposures, metaUro enables users to examine multiple exposures simultaneously, facilitating the identification of consistent associations, conflicting findings, and understudied relationships.

Table 1: Comparison with existing LLM-assisted evidence-synthesis approaches.

Approach	**Primary product**	**Cost/access burden to user**	**Large-scale/on-demand**
TrialMind [18]	Review workflow	Requires API access	User-specified individual reviews
meta-pipe [19]	System description; no validation data reported	Requires software/API setup	User-specified individual reviews
otto-SR [20]	Commercial tool	Customized commercial service	User-specified individual reviews
LatteReview [21]	Multi-agent review workflow	Requires technical setup and paid LLM APIs	User-specified individual reviews
Elicit [22]	Commercial tool	Subscription required for systematic reviews	User-specified individual reviews
metaUro	**Web platform**	**Freely accessible; no API key, software installation, or computing required**	**Continuously updated, large-scale, on-demand reviews**

In this paper, we present metaUro, which applies an automated LLM framework to investigate the associations between a broad range of nutritional exposures and the incidence of bladder, kidney and renal pelvis, and prostate cancers, including histological subtypes and aggressive prostate cancer. Section 2 describes the methods, Section 3.1 compares metaUro with two published meta-analyses, Section 3.2 compares associations across exposures and urologic cancers, Section 4 summarizes findings and limitations and discusses future directions.

2 Methods

metaUro adapts the nutritional evidence-synthesis framework of MetaFemina [23] to urologic cancer incidence, combining literature identification, model-assisted screening and extraction, principled appraisal, and quantitative synthesis in one web application. The exposure catalogues are based on fact sheets from the National Institutes of Health Office of Dietary Supplements [24], covering more than four hundred dietary exposures, including exposures with hierarchical relationships (for example, oleic acid is a subexposure of omega-9 fatty acids; Supplemental Table S.1). Each exposure is searched independently, but users can choose whether to include subexposure evidence in analyses of parent exposures, with inclusion as the default. By default, we include studies based on both dietary intake (food frequency questionnaires, dietary recalls, diet records, supplement intake reports, self-reported intake, and dietary patterns) and biospecimen measures (blood, serum, plasma, urine, saliva, tissue, fecal samples, metabolomics profiles, biomarker concentrations, or similar biological measurements), while users can specify the analysis based on the exposure measurement types. The disease catalog covers bladder cancer (urothelial carcinoma, squamous cell carcinoma, adenocarcinoma, and small cell carcinoma); kidney and renal pelvis cancer (renal cell carcinoma and its clear cell, papillary, and chromophobe subtypes; urothelial carcinoma of the renal pelvis; and renal sarcoma);

and prostate cancer, including an aggressive disease subgroup. For prostate cancer, we additionally conduct analyses excluding estimates specific to aggressive disease while retaining overall prostate cancer estimates, including those from studies covering both aggressive and nonaggressive types. We do not define a separate nonaggressive disease cohort. metaUro focuses on cancer incidence; prognostic endpoints such as survival, mortality, recurrence, progression are out of scope.

2.1 Evidence extraction framework

As summarized in Figure 1, we implemented a cost-efficient multi-level screening strategy that balances sensitivity, specificity, and computational efficiency. We first search PubMed using predefined aliases for each exposure and disease scope. Details of the search-query construction are provided in Supplemental Section S.1. In Round 1 (R1), a fast, low-cost LLM screens titles, abstracts, and Medical Subject Heading (MeSH) terms with high sensitivity for potential relevance to the prespecified exposure and disease. Only clear mismatches are excluded at this stage; uncertain cases and studies with incomplete quantitative reporting advance to Round 2 (R2). In R2, two independent, more capable LLMs conduct detailed eligibility assessment using prespecified criteria. Systematic reviews, meta-analyses, editorials, protocols, animal or other preclinical studies, and studies evaluating nonseparable joint exposures are excluded (exclusion counts by reason in the Claude–Gemini pipeline: Supplemental Table S.2). When key quantitative evidence is missing, or the JBI rating is below "Moderate" and the full text is available through the PubMed Central identifier (PMCID), two fast, low-cost LLMs independently re-evaluate each article using the full text (R3), respectively paired with the two R2 models. Agreement between the two pipelines is assessed in Supplemental Section S.2. Remaining disagreements after R3 were sent to a third, highly capable model, whose verdicts were applied only when backed by a verbatim quote validated against the source. Any remaining discrepancies were sent to a team member for adjudication. By reviewing discrepancies between the independent pipelines, we iteratively refine the prompt design, extraction logic, and screening workflow through minor adjustments. A detailed breakdown of the screening costs for both pipelines is provided in Supplemental Section S.3.

2.2 Principled evidence quality appraisal

The models applied study design-specific Joanna Briggs Institute (JBI) checklists (detailed criteria listed in Supplemental Table S.3) [25, 26, 27, 28]. For each question in the JBI critical appraisal checklist, if the relevant information is not reported in the article, then this question is graded as "Unclear". Based on the percentage of checklist questions answered "Yes", evidence quality is classified as Good ($> 80\%$), Moderate (51%–80%), or Fair ($<51\%$). Default analyses pool evidence with Moderate or Good rating; Fair-rated records remain available in the web platform with a gray layout indicating they are not included for calculating the pooled effect size. In the web platform, per-item JBI scores for each estimate are displayed when users hover over the quality appraisal.

2.3 Quantitative analysis and effect harmonization

Using the quantitative evidence stored in the backend, the platform automatically performs DerSimonian and Laird [29]'s random-effects meta-analysis, which accommodates between-study heterogeneity and returns a pooled effect estimate with the corresponding confidence interval (CI) and prediction interval (PI; computed when at least three estimates contributed) for a new study. Because relative risks (RRs) are more intuitive for communicating incidence associations and facilitate cross-study evidence synthesis [30, 31, 32], we convert hazard ratios (HRs), incidence rate ratios (IRRs), and odds ratios (ORs) to RRs for pooling the effect sizes. Conversion uses either a user-specified baseline risk or a risk obtained from the National Cancer Institute's Surveillance, Epidemiology, and End Results (SEER) Explorer, matched by cancer type, sex, race and ethnicity, starting age, and observation interval [33]. The default starting age is 60 years, with a 10-year observation interval. Let π_0 denote the baseline risk. ORs are converted using $\mathrm{RR} = \frac{\mathrm{OR}}{(1-\pi_0+\pi_0\mathrm{OR})}$, and HRs (under the proportional hazards assumption) and IRRs (under a constant-rate assumption) are converted using $\mathrm{RR} = \{1-(1-\pi_0)^{\mathrm{HR}}\}/\pi_0$. For rare subtypes whose incidence rate is not directly supported by SEER Explorer, and π_0 is not specified by the user, the platform uses the rare-disease approximation

OR ≈ RR ≈ HR. When a p-value is available but the confidence interval (CI) is not, we reconstruct the 95% CI using a Wald approximation on the log-ratio scale.

The web platform presents the heterogeneity statistic (I^2)[34], and offers leave-one-out analyses and Baujat plots [35] to identify outlier and influential studies. To inspect potential small-study effects, the platform generates funnel plots [36] and conducts Egger's regression test [37] when the number of included studies is ≥ 10. To support cautious interpretation, the platform generates diagnostic plots comparing heterogeneity, quantified by I^2, with the meta-analytic effect sizes (Supplemental Figure S.1) and evidence of publication bias, quantified by ($\log_{10}$)-transformed Egger's test p-values (Supplemental Figure S.2). Exposure–outcome pairs in the corner regions of these plots indicate findings with substantial between-study heterogeneity and either large meta-analytic effect sizes or stronger evidence of potential selective reporting, suggesting that these results should be interpreted with additional caution.

2.4 Interactive platform, result summary, automated sample size calculation, multilingual support, and monthly updates

The web platform allows users to sort studies by published year, sample size, and effect size; inspect study-level metadata with supporting excerpts from the source articles in an expandable panel; restrict analyses by design type, JBI rated quality, effect measurement type, and case count; select and deselect specific evidence for computing the pooled effect size; and export results. When multiple articles report findings from the same cohort, users have the option to retain the most recently published result eligible for pooling to avoid double-counting evidence from overlapping study populations. For each exposure–outcome combination, the platform automatically generates a downloadable PRISMA flow diagram [38] summarizing the numbers of records identified, screened, excluded at each review stage with reasons, and ultimately included.

Following the large-scale meta-analyses, we summarize findings across exposures for each disease outcome in forest plots as in Figure 2. Additionally, building on the pooled effect size, we provide literature-informed sample size calculations to support study planning, helping researchers quickly assess study feasibility, prioritize research questions, and make informed decisions before formally consulting a statistician. Users can specify the study design, target type I error rate, statistical power, test sidedness, and number of study arms. The formulas used for sample size calculations are provided in Supplemental Section S.4. To support sample size planning for new studies, the Sample Size Summary page reports the total sample size required to detect the pooled RR for each exposure supported by at least two studies. Supplemental Figure S.15 illustrates this functionality for kidney and renal pelvis cancer.

The interface supports five languages: English, Simplified Chinese, Traditional Chinese, Korean, and Dutch, with summary PDFs available in each to facilitate access by researchers worldwide. A monthly GitHub Actions update will capture newly published evidence, and team members will be assigned to adjudicate conflicting evidence from independent extraction pipelines. Discrepancies will be systematically reviewed to identify potential issues in prompt design, extraction logic, or the screening workflow. Confirmed problems will then be corrected and incorporated into subsequent platform updates.

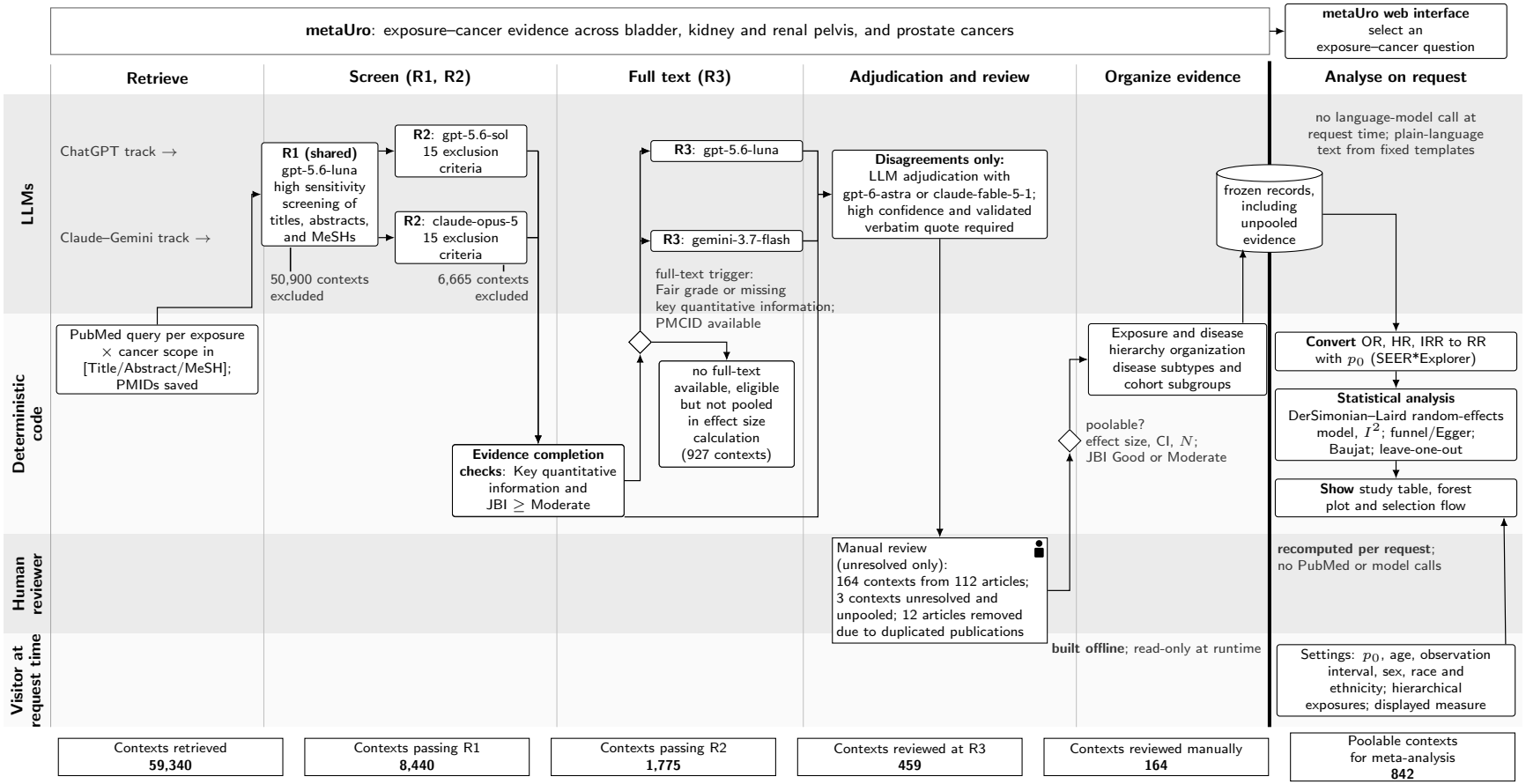


Figure 1: METAURO workflow. Columns show processing stages, and rows show the responsible actors. Evidence assembly to the left of the bold vertical line occurs offline, with outputs stored as write-protected, hash-listed files. Analyses to the right are recomputed for each request, except published summaries, which are rebuilt when their inputs change. The ChatGPT and Claude–Gemini tracks share round 1. Counts refer to article–exposure–disease contexts; an article may contribute to more than one context. R2 exclusions and R3 deferrals are those of the Claude–Gemini track. Records on which the two tracks agree, or whose disagreement the adjudicating model resolves, go directly to the pooling check. Abbreviations: R1–R3, screening rounds 1–3; p_0, baseline risk.

3 Results

3.1 Benchmarking metaUro against published evidence syntheses

In this subsection, we compare metaUro with two published meta-analyses of prospective studies: Qi et al. [39] on alcohol consumption and renal cell carcinoma (RCC), and Balali et al. [40] on dietary lycopene and prostate cancer. Because a corresponding 10-year risk estimate for renal cell carcinoma (RCC) was unavailable from SEER Explorer, metaUro pooled RCC effect estimates using the rare-disease approximation. For prostate cancer, metaUro used the default SEER Explorer-based 10-year risk of 5.45% for 60-year-old men of all races and ethnicities. Supplemental Table S.4 summarizes the pooled estimates from both comparisons, including analyses restricted to prospective designs and to one report per cohort.

3.1.1 Example 1: alcohol consumption and RCC

Table 2: Prospective studies on alcohol consumption and renal cell carcinoma included in Qi et al. [39]

Authors	Location	Sample size/cases	Exposure assessment and study name	Exposure contrast	Effect estimate (95% CI)
Nicodemus et al. [41]	USA	34,637/124	Mailed questionnaire (IWHS)	≥3.0 g/day vs nondrinkers	RR 0.52 (0.29–0.92)
Mahabir et al. [42]	Finland	27,111/195	Validated baseline questionnaire (ATBC)	Highest vs lowest quartile	RR 0.53 (0.34–0.83)
Rashidkhani et al. [43]	Sweden	59,237/132	FFQ (SMC)	>4.3 vs <2.5 g/day	RR 0.71 (0.42–1.19)
Lee et al. [44]	USA	136,587/248	Semiquantitative FFQ repeated every 2–4 years (NHS, HPFS)	≥15 g/day vs nondrinkers	RR 0.66 (0.43–1.00)
Allen et al. [45]	UK	1,280,296/1,141	Questionnaire on beverage type and amount (MWS)	Highest vs lowest alcohol intake category	RR 0.66 (0.48–0.92)
Wilson et al. [46]	Finland	27,111/228	Diet history questionnaire (ATBC)	Highest vs lowest alcohol intake category	RR 0.90 (0.50–1.60)
Lew et al. [47]	USA	492,187/1,814	FFQ (NIH-AARP)	≥30 vs >0 to <5 g/day	RR 0.69 (0.58–0.82)
Allen et al. [48]	UK	779,369/588	Beverage questionnaire approximately 3 years after recruitment (MWS)	≥2 vs <1 drink/day	RR 0.76 (0.61–0.96)
Macleod et al. [49]	USA	77,260/249	120-item FFQ (VITAL)	Highest vs lowest alcohol intake category	HR 0.69 (0.43–1.11)
Karami et al. [50]	USA	107,998/408	Questionnaire (PLCO)	>9.75 g/day vs non-drinkers	HR 0.67 (0.50–0.89)
Wozniak et al. [51]	Europe	477,325/931	Country-specific validated dietary questionnaires; baseline and lifetime intake (EPIC)	Highest vs light-drinker category	HR 0.94 (0.61–1.44)
Minami et al. [52]	Japan	105,663/340	Validated self-administered FFQ (JPHC)	Highest vs lowest alcohol intake category	HR 0.73 (0.44–1.22)

Abbreviations: ATBC, Alpha-Tocopherol, Beta-Carotene Cancer Prevention Study; EPIC, European Prospective Investigation into Cancer and Nutrition; FFQ, food-frequency questionnaire; HPFS, Health Professionals Follow-up Study; IWHS, Iowa Women's Health Study; JPHC, Japan Public Health Center–based Prospective Study; MWS, Million Women Study; NHS, Nurses' Health Study; NIH-AARP, National Institutes of Health–AARP Diet and Health Study; PLCO, Prostate, Lung, Colorectal, and Ovarian Cancer Screening Trial; SMC, Swedish Mammography Cohort; VITAL, VITamins And Lifestyle Study. *Note:* Gray shading indicates studies included in the previous meta-analysis but excluded from pooling in the corresponding metaUro analysis. Effect estimates are those used in the previous review, with the reported effect measure retained where known.

We here provide a guide to the metaUro interface and compare our outcome with that of Qi et al. [39] from a random-effects model, which included prospective studies through April 1, 2023, and reported a pooled RR of 0.70 (95% CI: 0.64, 0.77; $I^2 = 0\%$) for the highest versus the lowest intake [39].

First, the user chooses a disease scope (here, RCC) and a nutritional exposure from the drop-down list (here, alcohol; Supplemental Figure S.3). Clicking Analyze Evidence loads the saved results without new searches or LLM calls. The Study selection page provides the PRISMA plot showing the searching and screening workflow. For alcohol and RCC, 243 articles were fetched; 155 were excluded at the first screen and 50 after detailed assessment. The most common exclusion reasons are review or meta-analysis articles (26) and genotype-specific estimates (13). Of the 38 retained articles, 19 entered the pooled result, contributing 21 estimates. The remaining 19 were eligible but not pooled: 13 for missing key quantitative information and 6 for JBI quality below the pooling threshold (Supplemental Figure S.4).

For the studies that participated in the final effect size calculation, metaUro presents a forest plot (Supplemental Figure S.5) and the Extracted Studies table (Supplemental Figure S.6). Expanding a row displays the quotations from the article that support each extracted value (Supplemental Figure S.6). We adopt a crowdsourced verification mechanism. Users can validate extracted information by clicking a "Verified" check mark on the web platform, providing an additional layer of evidence reliability for future users. Users may also flag potentially problematic studies. Studies receiving two flags will trigger an email alert to the research team. Confirmed problems will then be corrected and incorporated into subsequent platform updates.

The pooled result is shown with an automatically generated plain-language interpretation (Supplemental Figure S.7). The 21 estimates from 19 studies gave RR=0.72 (95% CI: 0.65–0.79), suggesting that alcohol is associated with a lower risk of RCC. However, this association should be interpreted with caution, as there was substantial between-study heterogeneity ($\mathbf{I^2}$ = 52.61%), reflecting considerable variability across studies. metaUro also directly supports sample size planning. Using the unrounded pooled alcohol effect, a case–control study with 50% of controls in the higher-alcohol-intake group, a 1:1 case–control ratio, targeting two-sided $\alpha = 0.05$, and 80% power would require 572 cases and 572 controls.

Egger's test indicated asymmetry in the funnel plot (P = 0.0019; Supplemental Figure S.8). In the influence panel (Supplemental Figure S.9), the Baujat plot suggested Floud et al. [53]'s 2023 Million Women Study being the most influential estimate and a case–control study from Antwi et al. [54] as the largest contributor to heterogeneity. Leave-one-out estimates ranged from 0.70 to 0.75.

metaUro retrieved all 12 articles Qi et al. [39] included (Table 2). Among studies absent from the pooled results of metaUro, Nicodemus et al. [41] was a study in kidney cancer and was included in our kidney cancer analysis because RCC is a subtype of kidney cancer. Allen et al. [45] reported only a trend p-value for RCC in its abstract, and Wozniak et al. [51] reported estimates only for all renal cancers combined, which were included in our kidney and renal pelvis cancer analysis. Wilson et al. [46] focused on fish, vitamin D, and flavonoids, with alcohol included only as an adjustment covariate and no relevant quantitative effect estimates available from the abstract. Mahabir et al. [42] was not pooled because its abstract did not report the total sample size, which metaUro requires for pooling. The analysis also included 12 articles not included in the original review (Supplemental Table S.5). Of these, 6 were case–control studies that did not meet the review's design criteria; and 2 were eligible prospective studies from the Multiethnic Cohort [55] and the Netherlands Cohort Study [56] published before the review's literature search.

3.1.2 Example 2: lycopene and prostate cancer

Balali et al. [40] analyzed dietary and blood lycopene separately, reporting fixed-effect pooled estimates for prostate cancer of 0.95 (95% CI: 0.90–1.00; $I^2 = 16\%$) and 0.89 (95% CI: 0.80–0.98; $I^2 = 18\%$), respectively. Here, we focus our comparison on dietary lycopene intake (Table 3). metaUro pooled 13 estimates from 13 reports and yielded an overall effect estimate of 0.87 (95% CI: 0.78–0.98), suggesting an inverse association between lycopene intake and prostate cancer risk. However, substantial between-study heterogeneity ($I^2 = 69.84\%$) warrants cautious interpretation.

Using this pooled estimate, metaUro also provides sample size calculations. Assuming a starting age of 60 years and 10 years of follow-up, the reference cumulative incidence risk of prostate cancer is 5.453%. For a prospective two-arm study with equal allocation, a two-sided significance level of $\alpha = 0.05$, and 80% power to detect the pooled RR of 0.87 (unrounded in the calculation), approximately 32,604 participants (16,302 per group) would be required, corresponding to approximately 1,667 expected prostate cancer cases (Supplemental Figure S.10).

Among studies not pooled by metaUro, Schuurman et al. [57], Kirsh et al. [58], and Agalliu et al. [59] did not report lycopene-specific effect estimates in their abstracts, and their full texts were not available in PubMed Central. Kristal et al. [60] reported no lycopene-specific estimate in its abstract and was excluded at screening. Although Fraser et al. [61] reported estimates for tomato intake, tomato consumption is not equivalent to total lycopene intake, as lycopene is also found in other foods, including watermelon, papaya, and red bell peppers. Therefore, we did not substitute estimates for tomato intake for those of lycopene intake. Instead, metaUro synthesizes evidence on tomato intake separately. Additionally, metaUro pooled 10 reports not included by Balali et al. [40]: 5 prospective and 5 case–control studies (Supplemental Table S.6). Among these, Ge et al. [62] was published online on May 6, 2023, before the review's search cutoff,

whereas López-Solís et al. [63] was published after the cutoff. These findings highlight the sensitivity of the automated approach and the value of monthly updates in incorporating newly published evidence.

Table 3: Studies on dietary lycopene intake and prostate cancer included in Balali et al. [40]

Authors	Location	Sample size/Cases	Design	Exposure and assessment	Exposure contrast	Effect estimate (95% CI)
Schuurman et al. [57][a]	Netherlands	58,279/642	Case-cohort (NLCS)	150-item semiquantitative FFQ; dietary lycopene	Highest vs lowest quintile	RR 0.98 (0.71–1.34)
Kirsh et al. [58][a]	USA	29,361/1,338	Prospective cohort (PLCO)	137-item FFQ; dietary lycopene	Highest vs lowest quintile	RR 0.95 (0.79–1.13)
Kristal et al. [60][b]	USA, Canada	9,559/1,703	Prospective cohort of trial participants (PCPT)	FFQ; dietary lycopene	>10.92 vs <3.99 mg/day	RR 1.05 (0.90–1.21) for Gleason 2–7; RR 1.32 (0.76–2.31) for Gleason 8–10
Agalliu et al. [59][a]	Canada	2,525/661	Case-cohort (CSDLH)	FFQ; dietary lycopene	Highest vs lowest quintile	HR 0.82 (0.61–1.10)
Zu et al. [64]	USA	49,898/5,728	Prospective cohort (HPFS)	Semiquantitative FFQ, repeated; dietary lycopene	10.14–115.01 vs 0–3.68 mg/day	HR 0.91 (0.84–1.00)
Park et al. [65]	USA	75,216/7,115	Prospective cohort (MEC)	FFQ; dietary lycopene	≥2.02 vs <0.75 mg/1,000 kcal	RR 1.10 (0.93–1.31)
Lane et al. [66]	UK	5,245/1,717	Nested case–control (four cohorts and a trial-based case–control study)	Multiple-day food diaries; dietary lycopene	>2.14 vs <0.35 mg/day	OR 0.85 (0.67–1.07)
Fraser et al. [61][a]	USA	27,934/1,226	Prospective cohort (AHS-2)	FFQ with 24-hour recall calibration; dietary lycopene	>10.22 vs <1.89 mg/day	HR 0.94 (0.80–1.12)

Abbreviations: AHS-2, Adventist Health Study-2; CSDLH, Canadian Study of Diet, Lifestyle, and Health; FFQ, food-frequency questionnaire; HPFS, Health Professionals Follow-up Study; MEC, Multiethnic Cohort; NLCS, Netherlands Cohort Study; PCPT, Prostate Cancer Prevention Trial; PLCO, Prostate, Lung, Colorectal, and Ovarian Cancer Screening Trial.
Note: Gray shading indicates studies included in the previous meta-analysis but not pooled in the corresponding metaUro analysis. Effect estimates are those used in the previous review. Superscript letters give the status in metaUro: a, eligible but not pooled because the abstract provided no usable lycopene estimate and no full text was available in PubMed Central; b, excluded at screening because the abstract reported no lycopene-specific estimate.

3.2 Comparative analysis across exposures and urologic cancers

For each disease scope, the Summary page of metaUro gathers the pooled estimates of all exposures in forest plots that show every exposure with at least two pooled estimates, separately for pooled RRs above and below 1.00 and whether or not the 95% CI excludes 1; companion plots restrict the evidence to dietary intake and relate heterogeneity to the pooled RR and to Egger's test. By default, and in the results below, evidence for a subexposure also contributes to its parent exposure; for example, estimates for α-linolenic acid count toward ω-3 fatty acids. At least one poolable estimate was available for 102 exposures for prostate cancer, 59 for bladder cancer and 52 for kidney and renal pelvis cancer.

3.2.1 Exposures associated with prostate cancer

Of the 63 exposures with at least two estimates for prostate cancer, 32 had pooled RRs above 1.00 and 31 below, and the 95% CI excluded 1 for 9 and 13 of them, respectively. Statistically significant positive associations (Figure 2) were observed for glucosamine, 1.06 (95% CI: 1.01, 1.11); vitamin D, 1.13 (1.07, 1.19); processed meat, 1.13 (1.05, 1.21); calcium, 1.18 (1.10, 1.28); alcohol, 1.20 (1.10, 1.30); dairy, 1.22 (1.12, 1.34); α-linolenic acid, 1.31 (1.08, 1.58); monounsaturated fat, 1.33 (1.14, 1.56); and red meat, 1.60 (1.22, 2.10). Statistically significant negative associations (Figure 3) were observed for soy, 0.60 (0.41, 0.89); green tea, 0.62 (0.43, 0.89); genistein, 0.72 (0.54, 0.95); tomato, 0.74 (0.64, 0.85); tea, 0.75 (0.64, 0.89); selenium, 0.77 (0.67, 0.89); and lycopene, 0.85 (0.78, 0.93), and for linoleic acid, vitamin K, legumes, vitamin B6, soluble fiber and plant-based diet (pooled RRs from 0.71 to 0.91). Alcohol, calcium, selenium and vitamin D each pooled 28 or 29 estimates.

Restricting the analysis to dietary-intake evidence changed several results. The associations with vitamin D, 1.34 (0.90, 2.01; 3 estimates), selenium, 1.02 (0.93, 1.12; 10 estimates), and genistein were no longer statistically significant, and glucosamine, linoleic acid and vitamin K had fewer than two dietary-intake

estimates. Conversely, β-carotene, 0.79 (0.63, 0.98), and polyunsaturated fat, 1.21 (1.07, 1.36), became statistically significant, as did lutein and vitamin B12, each with two estimates.

Heterogeneity was common: of the 50 exposures with at least three estimates, 42 had I^2 of at least 50% and 24 of at least 75% (Supplemental Figure S.1), warranting investigation of study heterogeneity. Among the 19 exposures with at least 10 estimates, Egger's test gave $P < 0.05$ for 9, including alcohol, calcium, red and processed meat, selenium, lycopene and tomato (Supplemental Figure S.2); these associations should be interpreted with additional caution because small-study effects, including publication bias, may contribute.

3.2.2 Exposures associated with bladder cancer

For bladder cancer (Supplemental Figures S.11 and S.12), 24 of the 32 exposures with at least two estimates had pooled RRs below 1.00; red meat, 1.32 (1.11, 1.57), eggs and dietary and animal protein intake were positively associated, and 11 exposures inversely, including selenium, 0.73 (0.54, 0.97), folic acid and dietary fiber. Except for red meat, selenium and alcohol, these associations rested on two to five estimates, and 8 of the 11 inverse associations did not persist in dietary-intake evidence.

3.2.3 Exposures associated with kidney and renal pelvis cancer

For kidney and renal pelvis cancer (Supplemental Figures S.13 and S.14), red meat, 1.43 (1.13, 1.82), dairy, 1.57 (1.10, 2.25), and ultra-processed food were positively associated, and 11 of the 30 exposures inversely, the best supported being alcohol, 0.77 (0.70, 0.84; 29 estimates); most of these associations persisted in dietary-intake evidence. Subtype-specific evidence was sparse: renal cell carcinoma had 25 exposures with at least two estimates, urothelial carcinoma of the bladder one, and clear cell and papillary renal cell carcinoma none.

3.2.4 Comparison across urologic cancers

Figure 4 compares the 34 exposures with at least two estimates for at least two of the three cancers. Red meat showed the most consistent pattern, with statistically significant positive associations with prostate, 1.60 (1.22, 2.10), bladder, 1.32 (1.11, 1.57), and kidney and renal pelvis cancer, 1.43 (1.13, 1.82); processed meat pointed the same way for all three but was statistically significant only for prostate cancer. Selenium was inversely associated with all three, 0.77 (0.67, 0.89), 0.73 (0.54, 0.97) and 0.26 (0.08, 0.92), although the kidney estimate rested on two estimates. Dietary fiber, lycopene and tea had pooled RRs below 1.00 for all three cancers, each statistically significant for two, and folic acid was inversely associated with bladder and kidney cancers but not with prostate cancer. In contrast, alcohol was positively associated with prostate cancer, 1.20 (1.10, 1.30), but inversely with kidney and renal pelvis cancer, 0.77 (0.70, 0.84), and bladder cancer, 0.86 (0.74, 0.996). Dairy was positively associated with prostate and kidney cancers but not with bladder cancer, and the positive associations with vitamin D and calcium were confined to prostate cancer. The evidence was uneven: 19 of the 63 prostate cancer analyses pooled 10 or more estimates, whereas three-quarters of the urinary tract analyses pooled two to five.

4 Discussion

We developed metaUro, a freely accessible, LLM-assisted platform for synthesizing evidence on nutritional exposures and the incidence of bladder, kidney and renal pelvis, and prostate cancers. Compared with existing LLM-based automated meta-analysis tools, metaUro enables on-demand meta-analyses across a broad range of nutritional exposures. The platform also supports sample size calculations for future studies using pooled effect estimates.

In the two benchmark comparisons, metaUro retrieved all 12 and 8 articles included in the respective published meta-analyses, incorporated 7 and 3 of these into the pooled estimates, and additionally pooled 14 estimates from 12 reports and 10 estimates from 10 reports, respectively. Most articles retrieved but not pooled had no full text available through PubMed Central, and their abstracts lacked sufficient quantitative information for effect-size extraction. Other exclusions reflected a more conservative application of eligibility criteria, such as distinguishing renal cell carcinoma from the broader category of kidney cancer and

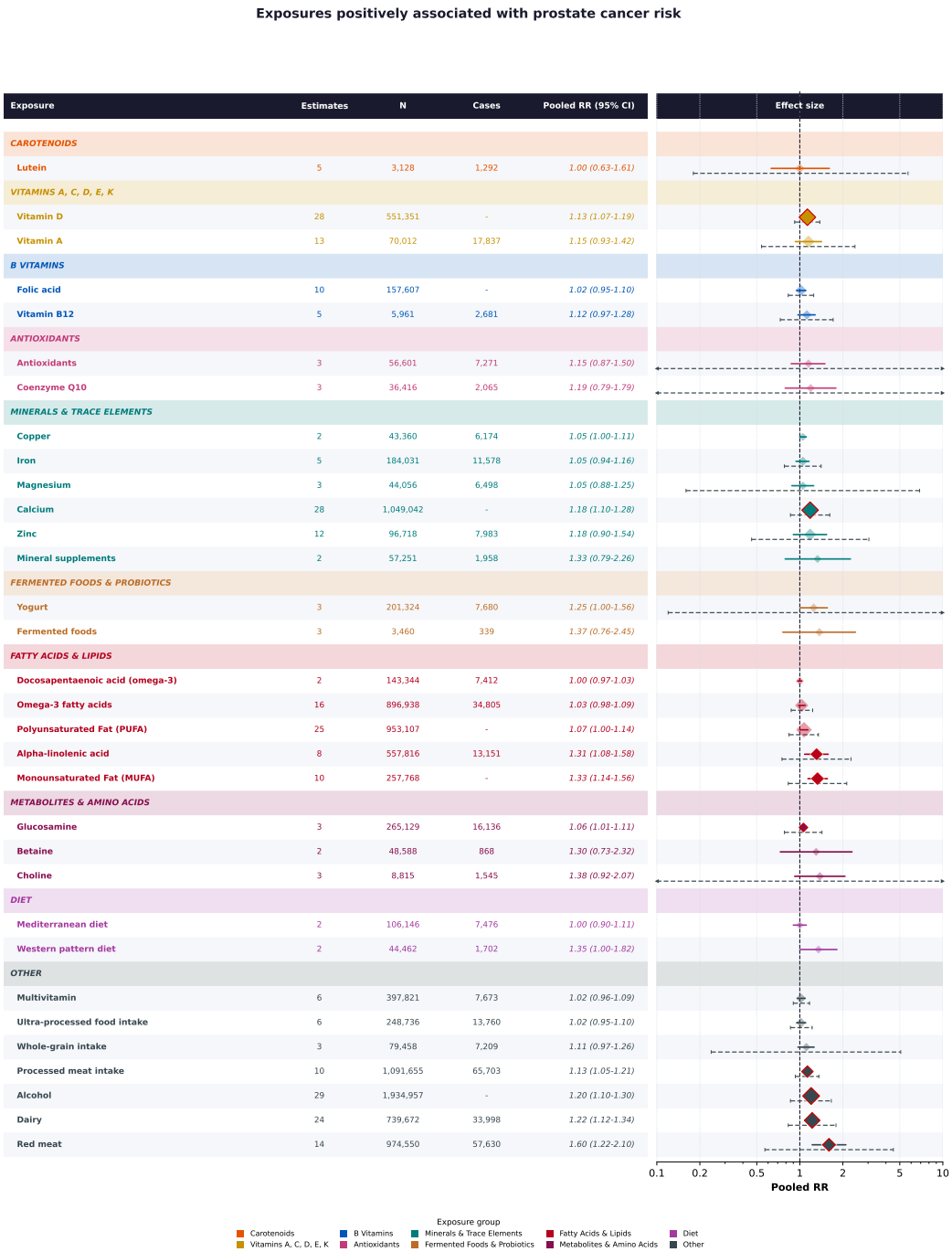


Figure 2: Summary forest plot of nutritional exposures with pooled risk ratios above 1.00 for prostate cancer in metaUro (at least two pooled estimates per exposure; default settings). Each row reports the number of pooled estimates (Estimates), total participants (N), and total cases (Cases), where available. Diamonds with a red outline mark 95% confidence intervals that exclude 1, and dashed lines show 95% prediction intervals. Colors denote exposure groups.

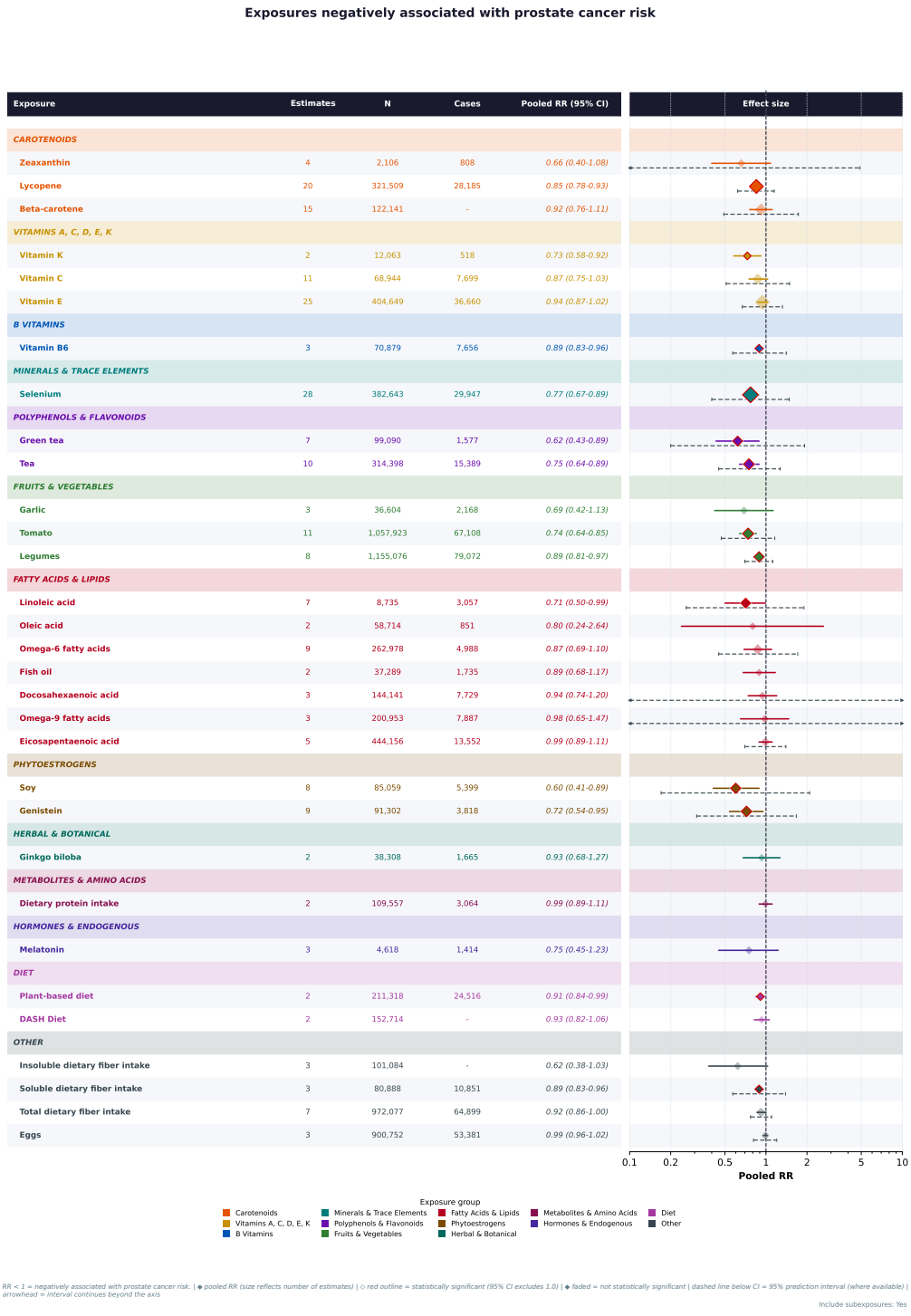


Figure 3: Summary forest plot of nutritional exposures with pooled risk ratios below 1.00 for prostate cancer in metaUro (at least two pooled estimates per exposure; default settings). Each row reports the number of pooled estimates (Estimates), total participants (N), and total cases (Cases), where available. Diamonds with a red outline mark 95% confidence intervals that exclude 1, and dashed lines show 95% prediction intervals. Colors denote exposure groups. Abbreviations: DASH, Dietary Approaches to Stop Hypertension.

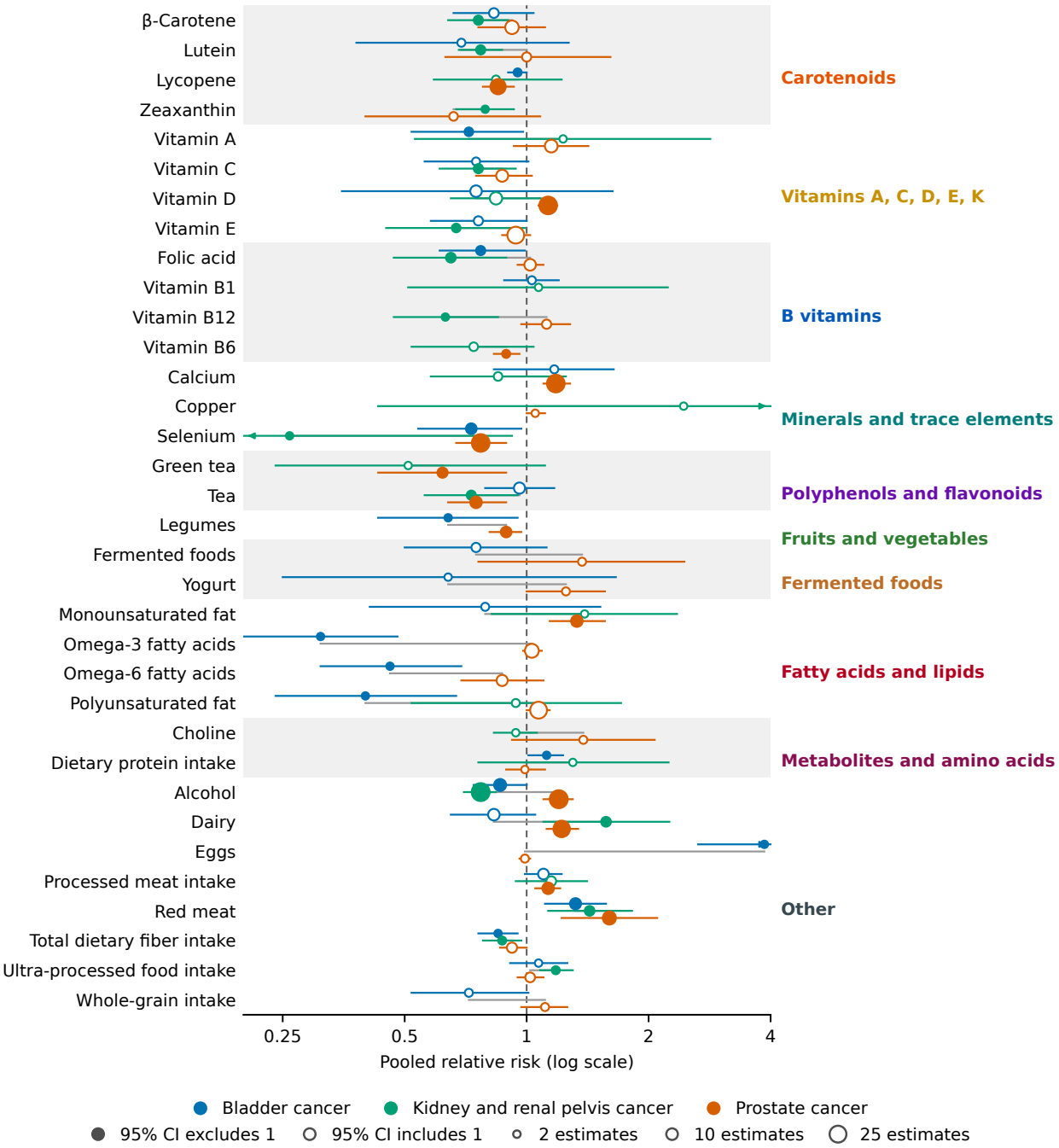


Figure 4: Pooled risk ratios of bladder, kidney and renal pelvis, and prostate cancer for the 34 exposures with at least two pooled estimates for at least two of the three cancers, in metaUro (default settings). Horizontal lines show 95% confidence intervals; filled markers indicate intervals that exclude 1 (judged on unrounded limits), and marker area reflects the number of pooled estimates. Arrows mark intervals that extend beyond the plotted range. Exposures are grouped as in metaUro's summary plots.

not treating lycopene intake as equivalent to tomato consumption. These decisions follow the prespecified exposure and outcome definitions.

The across-exposure summary showed patterns that single reviews do not readily reveal. For example, red meat was positively and selenium inversely associated with all three cancers, whereas alcohol was positively associated with prostate cancer but inversely with kidney and bladder cancers. Effect heterogeneity was common: 42 of the 50 prostate cancer analyses with at least three estimates had an I^2 of 50% or more, reflecting differences in study populations, exposure measurement and group categorization. A caveat in nutritional studies is that a single food supplies several nutrients, so the associations of a food and of its nutrients cannot be fully disentangled; dairy, for example, supplies calcium, protein and fat, and both dairy and calcium were positively associated with prostate cancer.

Nevertheless, metaUro has several limitations. First, automated screening and data extraction are susceptible to LLM misinterpretation and hallucination. We mitigate these risks through two independent LLM pipelines, automated adjudication supported by verbatim excerpts, and manual review by team members. Second, information extraction relies primarily on abstracts, because full texts are obtained only through PubMed Central. In the ChatGPT and Claude + Gemini pipelines, 747 and 717 articles required R3 full-text screening, of which 240 and 233 were completed, respectively. A subsequent search in Europe PMC [67] by PMID or title and first author's surname retrieved one additional article.

It is worth noting that metaUro is designed for evidence synthesis and hypothesis generation, not for generating causal claims. However, repeated associations, such as the associations with red meat and selenium, can help researchers identify exposures that merit targeted investigations.

Future work will focus on developing a public-facing interface for people without a biomedical or statistical background, harmonizing heterogeneous exposure measurements into common units, and validating automated extraction against more manually conducted meta-analyses. We will also extend the framework beyond disease incidence to prognosis endpoints, including recurrence, major adverse events, treatment response, and survival.

Funding

This research was funded by the National Science Foundation, grant number NSF DMS 2310955, and the Susan G. Komen Foundation.

Supplemental Materials

The Supplemental Materials provide query construction details and an example query of dairy intake and prostate cancer risk (Supplemental Section S.1), agreement assessment between LLM pipelines (Supplemental Section S.2), the computational costs of developing metaUro (Supplemental Section S.3), and automated sample size calculation details (Supplemental Section S.4).

S.1 Query construction and example query

Query construction. Searches used the Biopython interface to the National Center for Biotechnology Information (NCBI) E-utilities [68]: an ESearch call on the PubMed database returned identifiers and an EFetch call returned Extensible Markup Language (XML) metadata. Each search job paired one exposure with one disease scope, and its query had exactly two blocks joined by AND. The exposure block joined by OR every term of that exposure, each written as a double-quoted phrase restricted to [Title/Abstract]; exposures had no Medical Subject Headings (MeSH) clause, truncation or wildcard. For each searched disease scope (the 12 urinary scopes and prostate cancer), metaUro used OR to join the relevant disease terms; aggressive prostate cancer was not searched separately but derived from the prostate cancer evidence. An example query is shown below:

Example query.

```
(("Dairy"[Title/Abstract] OR "butter"[Title/Abstract] OR "cheese"[Title/Abstract] OR
"cream"[Title/Abstract] OR "cultured dairy"[Title/Abstract] OR "dairy products"[Title/Abstract] OR
"fermented milk products"[Title/Abstract] OR "kefir"[Title/Abstract] OR "milk"[Title/Abstract] OR
"yogurt"[Title/Abstract]) AND ("Prostatic Neoplasms"[MeSH Terms] OR "prostate
cancer"[Title/Abstract] OR "prostatic cancer"[Title/Abstract] OR "prostate
carcinoma"[Title/Abstract] OR "prostatic carcinoma"[Title/Abstract]))
```

S.2 Agreement assessment between two LLM pipelines

Across all urological cancer scopes, 8,440 article–exposure–disease contexts passed the shared first-round screening (an article can contribute more than one context). Comparing each pipeline's decisions after rounds 2 and 3, both pipelines excluded 6,560 contexts and retained 1,588, while 173 were retained only by the Claude–Gemini pipeline and 119 only by the ChatGPT pipeline, yielding 292 disagreements. (The Claude–Gemini pipeline excluded 6,665 contexts at round 2, the count shown in Figure 1, and 14 more at round 3.) Observed agreement was 96.5% (8,148 of 8,440). Cohen's $\kappa = (A_{\text{obs}} - A_{\text{exp}})/(1 - A_{\text{exp}})$ [69], where A_{obs} and A_{exp} denote observed and expected agreement, respectively, was 0.89. Positive agreement on retention, defined as twice the number of jointly retained contexts divided by that quantity plus disagreements, was 91.6%. These statistics measure agreement between model pipelines, not accuracy against an independent reference standard.

S.3 Computation cost

The ChatGPT pipeline cost an estimated US$442.85 based on public list prices, including US$430.72 for R2 (gpt-5.6-sol) and US$12.12 for R1 and R3 combined (gpt-5.6-luna). The Claude–Gemini pipeline reused R1 results and incurred US$157.55 for R2 (claude-opus-5) and US$12.97 for R3 (gemini-3.7-flash). Claude-fable-5-1 adjudication added US$34.08, bringing the total to US$647.45. Components and totals were rounded independently, so they may not sum exactly.

S.4 Automated sample size calculation details

Let ρ be the chosen risk ratio, α the significance level (default 0.05), s the number of sides of the test (2 by default; the in-page calculator also allows 1), $1 - \beta$ the power (default 80%), p_0 the baseline (reference-group) cumulative risk over the observation interval, and $z_\gamma = \Phi^{-1}(\gamma)$ the γ quantile of the standard normal

distribution. For a prospective comparison of two equal groups, with risks $\pi_A = p_0$ and $\pi_B = \rho p_0$ (requiring $\rho p_0 < 1$) and $\bar{\pi} = (\pi_A + \pi_B)/2$, the size of each group is

$$n = \left\lceil \frac{\left\{ z_{1-\alpha/s}\sqrt{2\bar{\pi}(1-\bar{\pi})} + z_{1-\beta}\sqrt{\pi_A(1-\pi_A) + \pi_B(1-\pi_B)} \right\}^2}{(\pi_B - \pi_A)^2} \right\rceil, \tag{1}$$

the total is $2n$, and the expected number of cases is $\lceil n(\pi_A + \pi_B) \rceil$. The in-page calculator also offers a single-arm design, in which the numerator becomes $\{z_{1-\alpha/s}\sqrt{\pi_A(1-\pi_A)} + z_{1-\beta}\sqrt{\pi_B(1-\pi_B)}\}^2$ with the same denominator, the total is n and the expected number of cases is $\lceil n\pi_B \rceil$. For an unmatched case–control study, the risk ratio is first converted to an OR, $\psi = \rho(1-p_0)/(1-\rho p_0)$ ($\psi = \rho$ when p_0 is missing); with exposure prevalence q_0 among controls (default 50%) and c controls per case (default 1), $q_1 = \psi q_0/(1 - q_0 + \psi q_0)$ and $\bar{q} = (q_1 + c q_0)/(1 + c)$, the numbers of cases and controls are

$$n_{\text{case}} = \left\lceil \frac{\left\{ z_{1-\alpha/s}\sqrt{(1+1/c)\,\bar{q}(1-\bar{q})} + z_{1-\beta}\sqrt{q_1(1-q_1) + q_0(1-q_0)/c} \right\}^2}{(q_1 - q_0)^2} \right\rceil, \qquad n_{\text{control}} = \lceil c\, n_{\text{case}} \rceil; \tag{2}$$

once the OR ψ is specified, the case–control calculation needs no further incidence input. Converting an RR to an OR uses p_0; when p_0 is unavailable, the calculator uses the rare-outcome approximation $\psi = \rho$.

Table S.1: Exposure hierarchy in metaUro (11 parents, 34 child exposures).

Parent exposure	Child exposures
Polyunsaturated fatty acids	Omega-3 and omega-6 fatty acids; their 12 single fatty acids (listed below); mead acid (an omega-9 fatty acid)
Omega-3 fatty acids	Alpha-linolenic, stearidonic, eicosatetraenoic (n-3), eicosapentaenoic, docosapentaenoic (n-3) and docosahexaenoic acids
Omega-6 fatty acids	Linoleic, gamma-linolenic, dihomo-gamma-linolenic, arachidonic, adrenic and docosapentaenoic (n-6) acids
Monounsaturated fatty acids	Omega-7 fatty acids; palmitoleic and cis-vaccenic acids; oleic, erucic, gondoic and nervonic acids
Omega-7 fatty acids	Palmitoleic and cis-vaccenic acids
Omega-9 fatty acids (no parent)	Oleic, erucic, gondoic, nervonic and mead acids
Collagen	Collagen types I, II, III, V and X
Mushrooms	Lion's mane
Vitamin B3	Nicotinamide; nicotinic acid
Vitamin B6	Pyridoxal 5'-phosphate
Vitamin K	Vitamins K1, K2 and K3

Notes: Each descendant is linked directly to every ancestor, so a parent's children are also all of its descendants. A child can have several parents (for example, docosahexaenoic acid sits under omega-3 fatty acids and under polyunsaturated fatty acids). Omega-9 fatty acids have no parent; each of their children maps to monounsaturated or polyunsaturated fatty acids directly.

Table S.2: Reasons for exclusion in R2 and R3 screenings.

Exclusion reason	Number of excluded contexts
Review or meta-analysis without an eligible primary estimate	3,265
Not an eligible primary study	403
Animal, laboratory, or preclinical study	845
Exposure outside the review scope	471 (4)
Target exposure inseparable from a mixture	242
Cancer outside the review scope	226
Outcome outside the review scope	96 (3)
Exposure assessed only after diagnosis	50
Benign or nonmalignant disease only	1
Mendelian randomization or genetic instruments	208
Genotype-specific effect or gene–exposure interaction	423
Genetic focus without a population-wide exposure estimate	204
Diagnostic or mechanistic research	138
No separable target incidence estimate	106 (7)
Duplicate or overlapping analysis at eligibility review	1
Total	6,679

Notes: Counts represent article–exposure–disease contexts excluded in the Claude–Gemini track by claude-opus-5 at round 2 or gemini-3.7-flash at round 3. Parenthetical values indicate the number excluded at round 3 and are included in the corresponding totals (e.g., 471 (4) = 467 + 4).

Table S.3: JBI critical-appraisal items used by metaUro, and the scoring rule.

Item number	Appraisal criterion
Cohort (11 items)	
1	Groups similar and recruited from the same population
2	Exposures measured similarly for both groups
3	Exposure measured validly and reliably
4	Confounders identified
5	Strategies for confounding stated
6	Participants free of the outcome at the start
7	Outcomes measured validly and reliably
8	Follow-up time reported and long enough
9	Follow-up complete, or losses described and explored
10	Strategies for incomplete follow-up used
11	Appropriate statistical analysis
Case–control (10 items)	
1	Groups comparable apart from the disease
2	Cases and controls matched appropriately
3	The same identification criteria for cases and controls
4	Exposure measured in a standard, valid and reliable way
5	Exposure measured the same way for cases and controls
6	Confounders identified
7	Strategies for confounding stated
8	Outcomes assessed in a standard, valid and reliable way
9	Exposure period long enough to be meaningful
10	Appropriate statistical analysis
Analytical cross-sectional (8 items)	
1	Inclusion criteria defined
2	Subjects and setting described
3	Exposure measured validly and reliably
4	Objective, standard criteria for the condition
5	Confounders identified
6	Strategies for confounding stated
7	Outcomes measured validly and reliably
8	Appropriate statistical analysis
Randomized controlled trial (13 items)	
1	True randomization
2	Allocation concealed
3	Groups similar at baseline
4	Participants blinded
5	Those delivering treatment blinded
6	Groups treated identically apart from the intervention
7	Outcome assessors blinded
8	Outcomes measured the same way in all groups

Continued on next page

Item number	Appraisal criterion
9	Outcomes measured reliably
10	Follow-up complete, or differences described
11	Participants analysed in the groups to which they were randomized
12	Appropriate statistical analysis
13	Appropriate trial design, with deviations accounted for
Scoring and operational categories	
Responses	Each item is answered Yes, No, Unclear or NA.
Score	$S = 100\,J_{\mathrm{Y}}/J$, where J_{Y} is the number of Yes answers and J is the number of items in the checklist. NA remains in the denominator.
Categories	Good, $S > 80$; Moderate, $51 \leq S \leq 80$; Fair, $S < 51$.
Default pooling	Requires Good or Moderate.

Notes: Abbreviations: NA, not applicable. Checklists: cohort [25], case–control [26], randomized controlled trial [27], analytical cross-sectional [28].

Table S.4: Pooled estimates from the published meta-analyses and from metaUro under default and restricted analysis settings.

Exposure and outcome	Published meta-analysis	metaUro, all designs	metaUro, prospective designs	metaUro, prospective, one report per cohort
Alcohol, RCC	0.70 (0.64, 0.77) 12; 0%	0.72 (0.65, 0.79) 21; 53%	0.77 (0.71, 0.83) 13; 22%	0.71 (0.64, 0.79) 9; 0%
Lycopene, prostate cancer				
Diet and biospecimen	–	0.84 (0.76, 0.92) 20; 72%	0.90 (0.82, 0.98) 13; 72%	–
Dietary intake	0.95 (0.90, 1.00) 8; 16%	0.87 (0.78, 0.98) 13; 70%	0.94 (0.84, 1.05) 8; 71%	0.98 (0.82, 1.16) 5; 77%
Biospecimen	0.89 (0.80, 0.98) 14; 18%	0.73 (0.55, 0.96) 6; 73%	0.70 (0.51, 0.95) 5; 78%	0.71 (0.50, 1.02) 4; 77%

Entries are RR (95% CI); number of estimates (of studies, for the published meta-analyses); I^2. One report per cohort: for alcohol, the Netherlands Cohort Study kept its 2008 report and the Million Women Study its 2011 report, and a pooling project of 12 cohorts was dropped; for dietary lycopene, the Health Professionals Follow-up Study kept its 2014 report; for biospecimen lycopene, a pooled analysis of 15 studies was dropped. Measurement type was taken from metaUro's records; the 20 combined estimates are 13 dietary, 6 biospecimen and 1 case–control estimate without a recorded measurement type, which enters the combined analysis only.

Table S.5: Estimates pooled in metaUro's analysis of alcohol consumption and renal cell carcinoma from reports not included in the comparator meta-analysis by Qi et al. [39].

Report	RR (95% CI)	Additional comment
Setiawan et al. [55] (MEC)	0.69 (0.49, 0.96)	Met the comparator's criteria
Schouten et al. [56] (NLCS)	0.69 (0.44, 1.07)	Met the comparator's criteria

Continued on next page

Report	RR (95% CI)	Additional comment
Floud et al. [53] (MWS)	0.88 (0.83, 0.94)	Published after the comparator's search; update of a pooled cohort
Lee et al. [70]	0.72 (0.60, 0.86)	Pooled analysis of 12 cohorts; overlaps cohorts already represented
Schouten et al. [71] (NLCS)	0.78 (0.48, 1.24)	Same cohort as Schouten et al. [56]; clear-cell RCC
van de Pol et al. [72] (NLCS)	0.77 (0.49, 1.23)	Same cohort as Schouten et al. [56]; RCC subtypes
Goodman et al. [73]	0.60 (0.40, 1.00)	
Parker et al. [74], women	0.50 (0.20, 0.90)	
Parker et al. [74], men	1.10 (0.80, 1.50)	
Greving et al. [75]	0.60 (0.40, 0.90)	
Pelucchi et al. [76]	0.70 (0.50, 0.97)	
Hu et al. [77], men	0.70 (0.50, 0.90)	
Hu et al. [77], women	0.70 (0.50, 0.90)	
Antwi et al. [54]	0.52 (0.42, 0.65)	

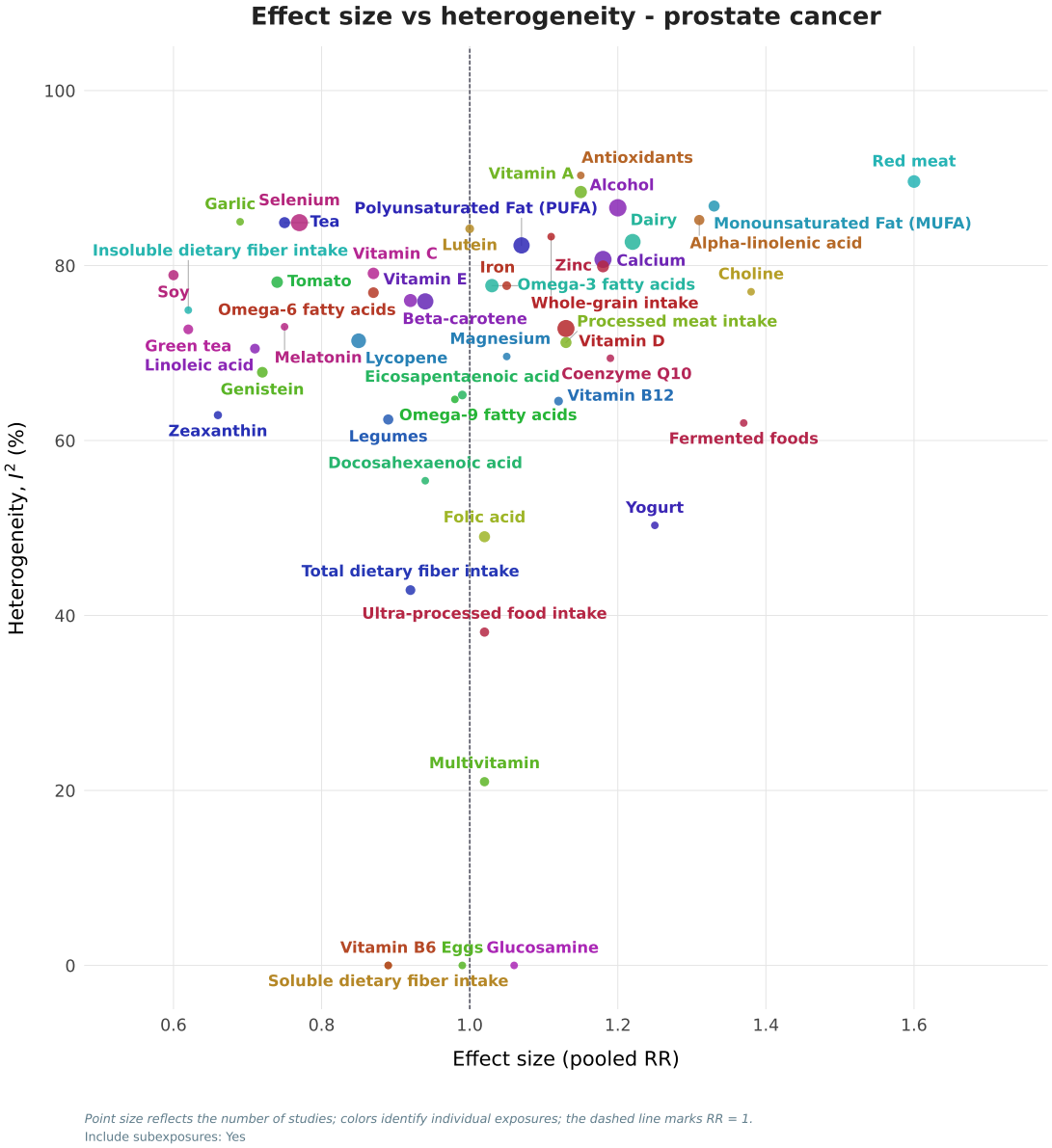
Notes: A total of 14 estimates from 12 reports (6 prospective and 8 retrospective estimates). Light pink shading identifies prospective studies that met Qi et al. [39]'s inclusion criteria before its literature search ended. Light blue shading identifies case–control studies, which were excluded by the comparator's study-design criteria. Unshaded rows represent other prospective studies. Abbreviations: MEC, Multiethnic Cohort; MWS, Million Women Study; NLCS, Netherlands Cohort Study.

Table S.6: Evidence pooled in METAURO's analysis of dietary lycopene intake and prostate cancer from reports not included by Balali et al. [40].

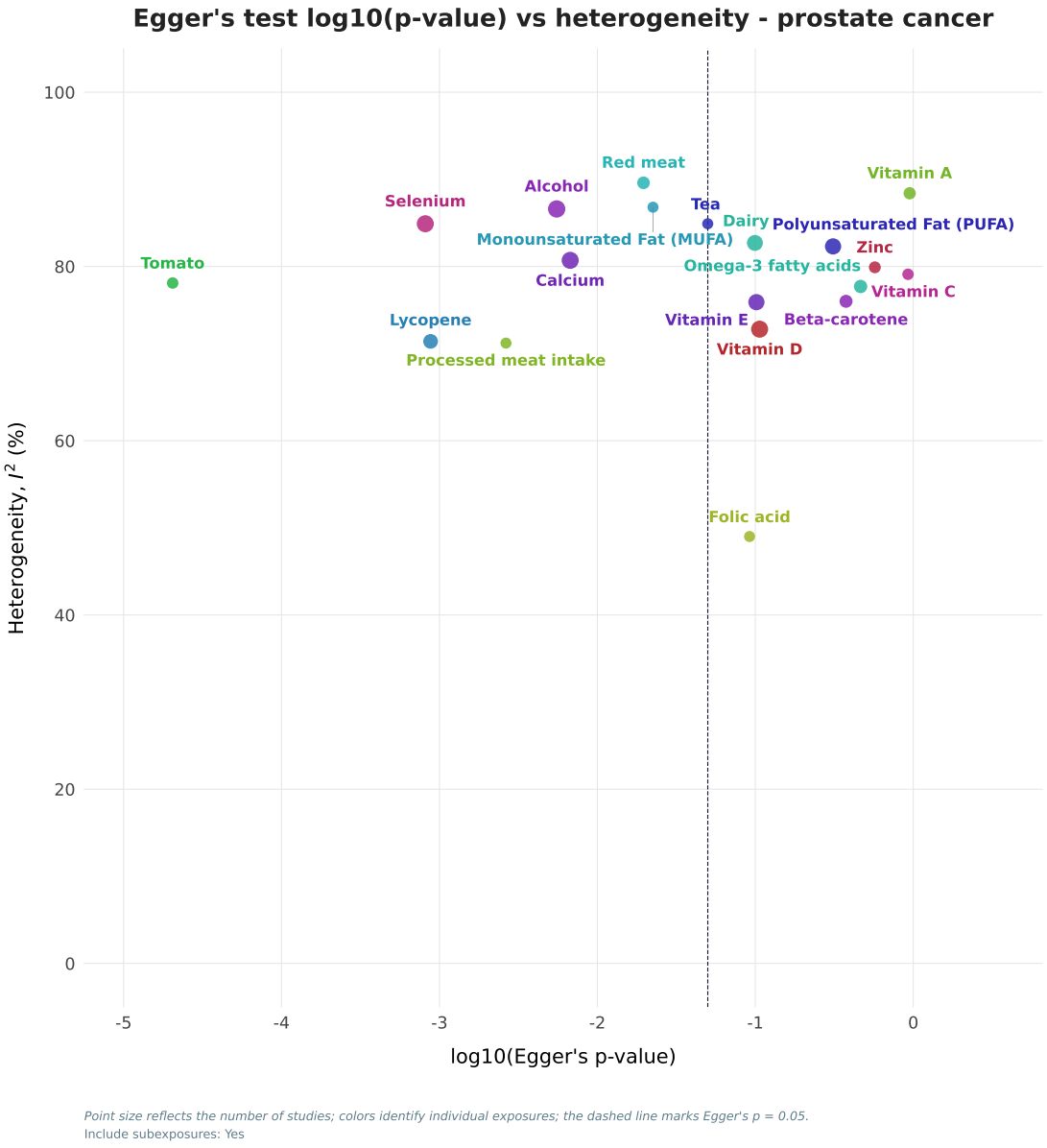
Report	RR (95% CI)	Additional comment
Ge et al. [62] (JPHC)	1.23 (1.04, 1.45)	Met the comparator's criteria
López-Solís et al. [63] (PREDIMED)	0.47 (0.23, 0.95)	Published after the comparator's search
Giovannucci et al. [78] (HPFS)	0.79 (0.64, 0.99)	Cohort already represented
Giovannucci et al. [79] (HPFS)	0.84 (0.73, 0.96)	Cohort already represented
Feng et al. [80] (HPFS)	1.01 (0.87, 1.17)	Lethal cancer after a negative biopsy; published after the comparator's search; cohort already represented
Key et al. [81]	0.99 (0.69, 1.42)	
Sanderson et al. [82]	0.72 (0.47, 1.08)	
McCann et al. [83]	0.63 (0.43, 0.92)	
Shahar et al. [84]	0.41 (0.17, 1.01)	
Van Hoang et al. [85]	0.47 (0.28, 0.78)	

Notes: A total of 10 dietary lycopene estimates from 10 reports (5 prospective and 5 retrospective). Light pink shading identifies prospective studies that met the comparator's inclusion criteria and were available before its literature search ended. Light blue shading identifies case–control studies, which were excluded by the comparator's study-design criteria. Unshaded rows represent other prospective studies. The comparator's search ended in July 2023. Abbreviations: HPFS, Health Professionals Follow-up Study; JPHC, Japan Public Health Center–based Prospective Study; PREDIMED, Prevención con Dieta Mediterránea.

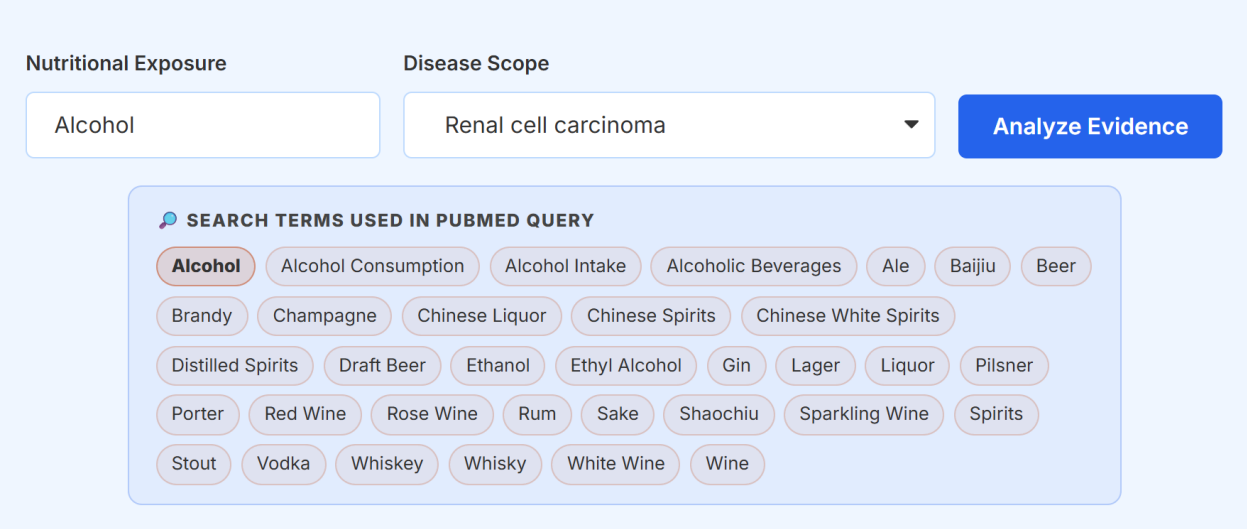
Supplemental Figures



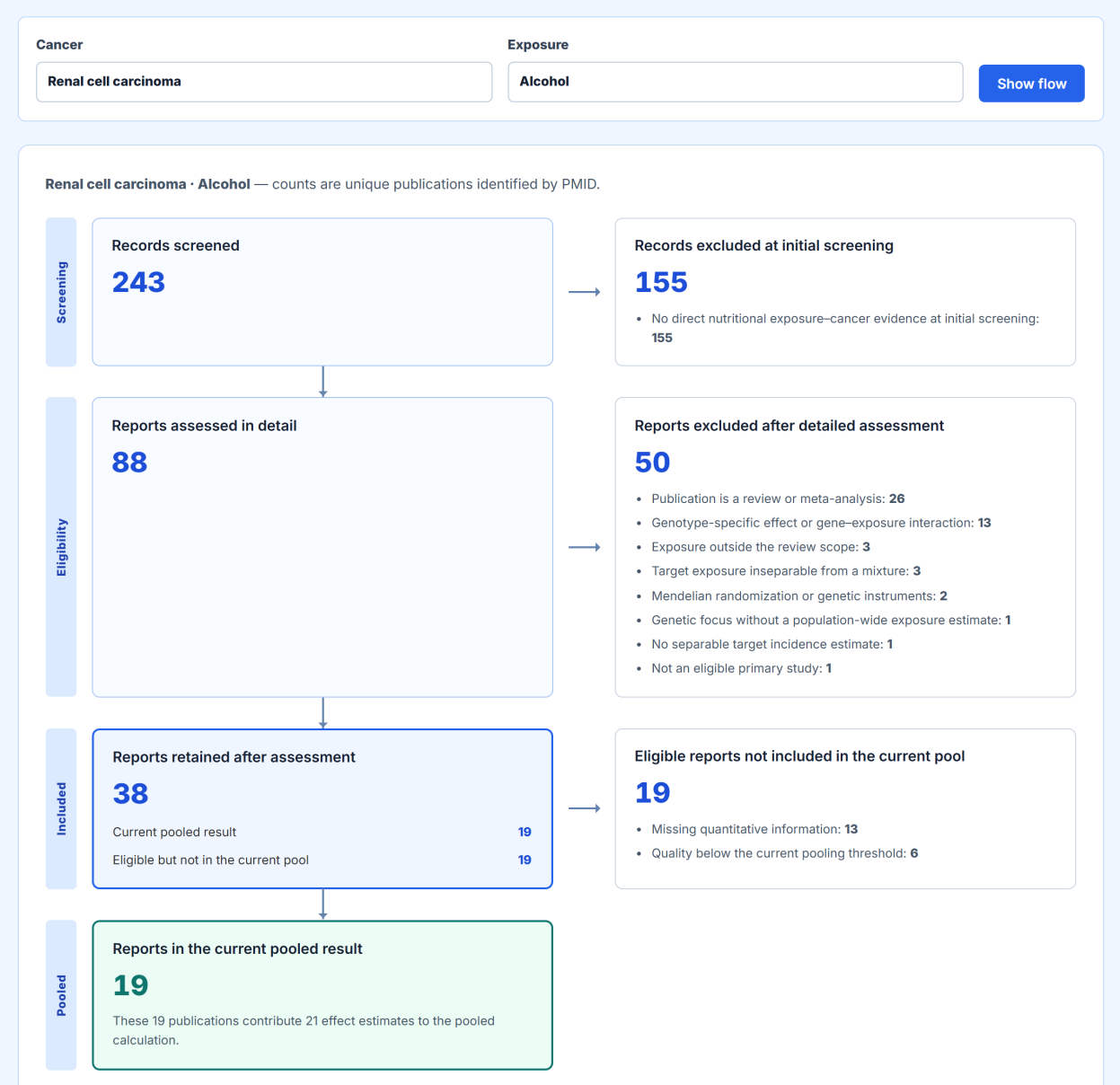
Supplemental Figure S.1: Heterogeneity (I^2) against the pooled risk ratio for the 50 exposures with at least three pooled estimates for prostate cancer, from MetaUro's Summary page (default settings). Point size reflects the number of pooled estimates; the dashed line marks a risk ratio of 1.



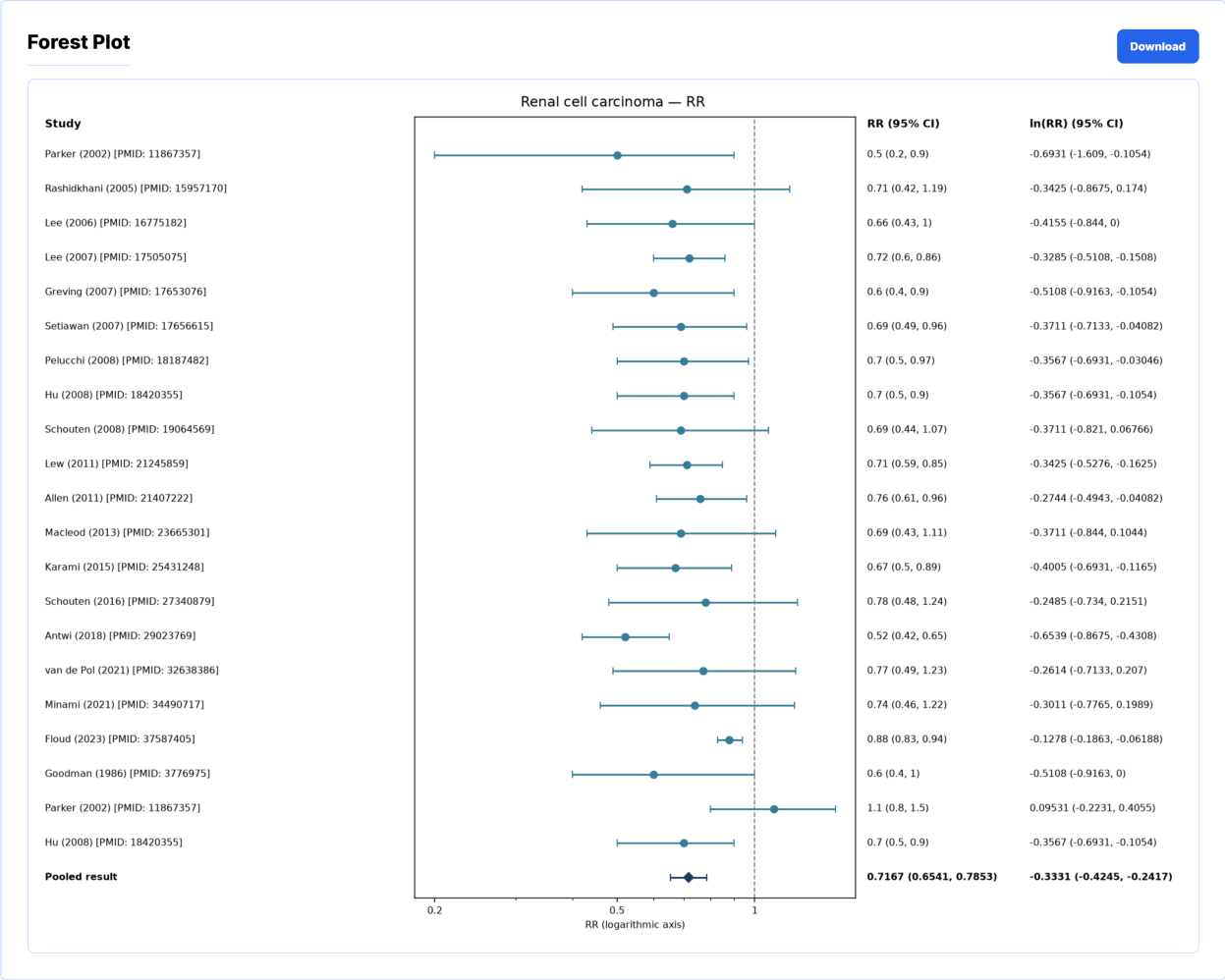
Supplemental Figure S.2: Heterogeneity (I^2) against $\log_{10}$ of the P value of Egger's test for the 19 exposures with at least 10 pooled estimates for prostate cancer (default settings). The dashed line marks $P = 0.05$; exposures to its left show evidence of small-study effects.



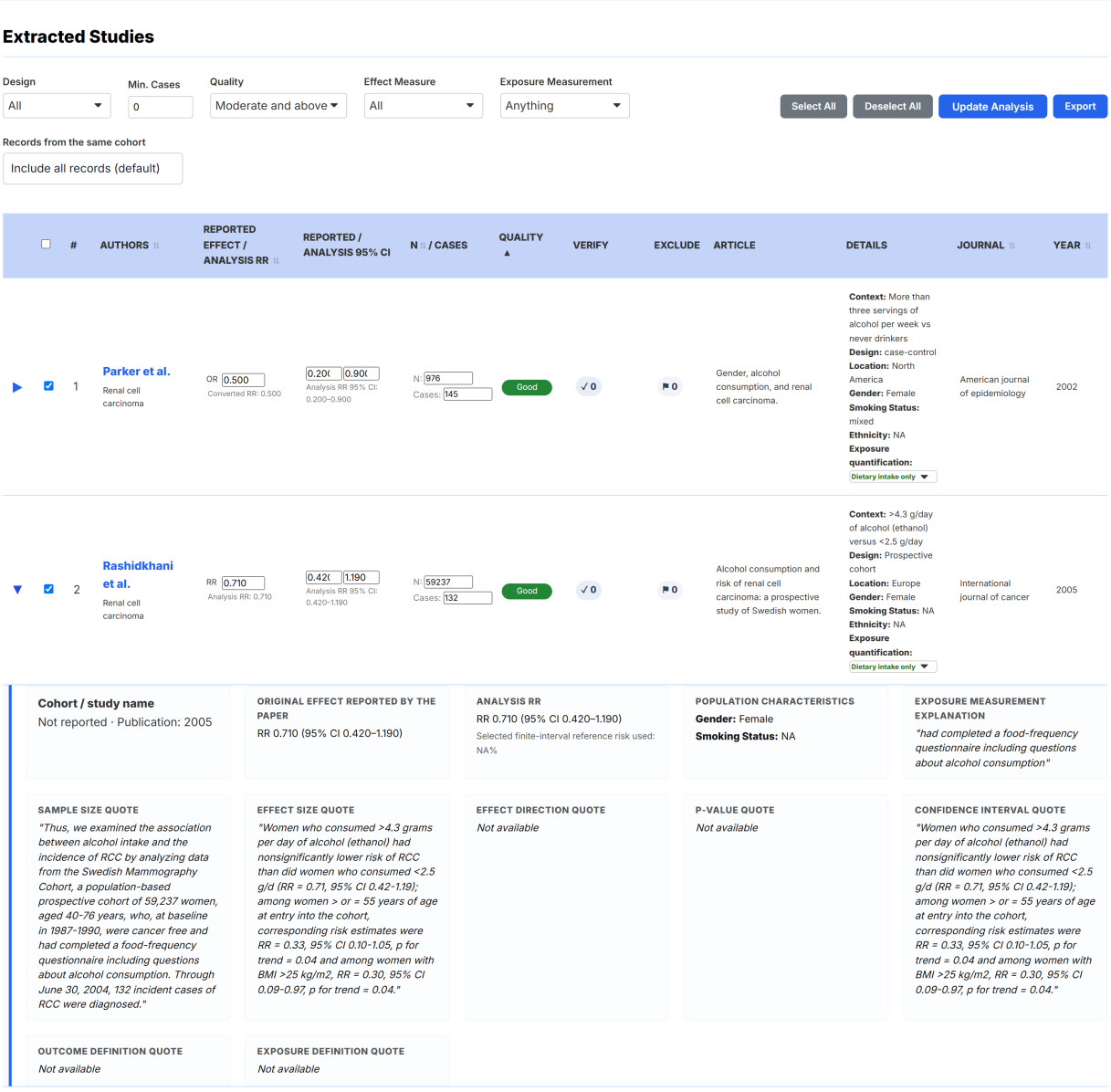
Supplemental Figure S.3: Selecting a disease scope and a nutritional exposure in METAURO (renal cell carcinoma and alcohol). The terms combined in the PubMed query for the selected exposure are listed below the selection.



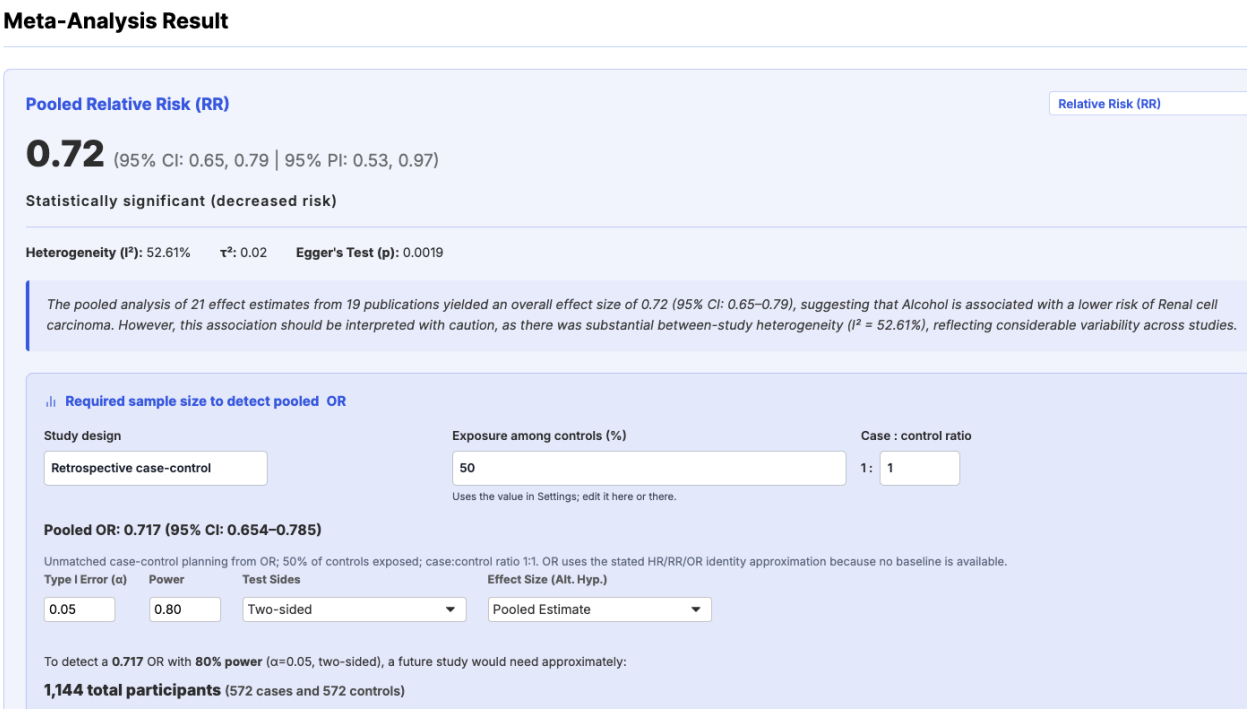
Supplemental Figure S.4: Study-selection flow for alcohol and renal cell carcinoma, with counts of unique publications (PubMed identifiers) and the recorded reasons for exclusion and for not pooling eligible reports.



Supplemental Figure S.5: Forest plot of the 21 estimates from 19 reports pooled for alcohol and renal cell carcinoma, with the random-effects pooled risk ratio (RR).



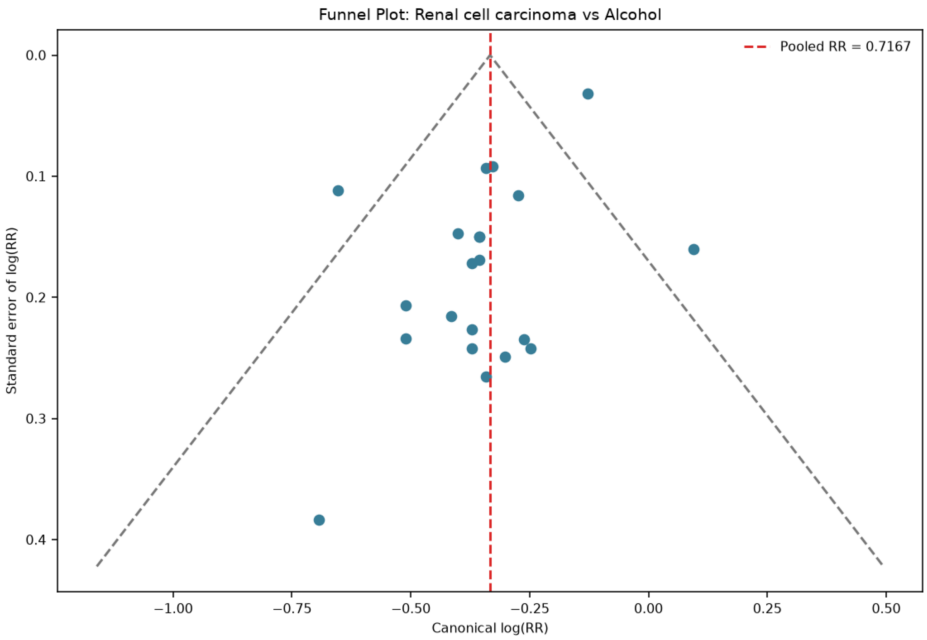
Supplemental Figure S.6: Extracted Studies table for alcohol and renal cell carcinoma, with its filters, selection controls and all columns. The second record is expanded to show the quotations from the article that support the extracted sample size and effect estimate.



Supplemental Figure S.7: Pooled RR for alcohol and renal cell carcinoma, with 95% confidence and prediction intervals, heterogeneity statistics, Egger's test, an automatically generated plain-language interpretation and the sample-size planner.

Publication Bias (Funnel Plot)

Download



How to read: Studies should ideally be distributed symmetrically within the white triangle. Asymmetry (clusters on one side) can indicate publication bias, where smaller studies with negative results remain unpublished.

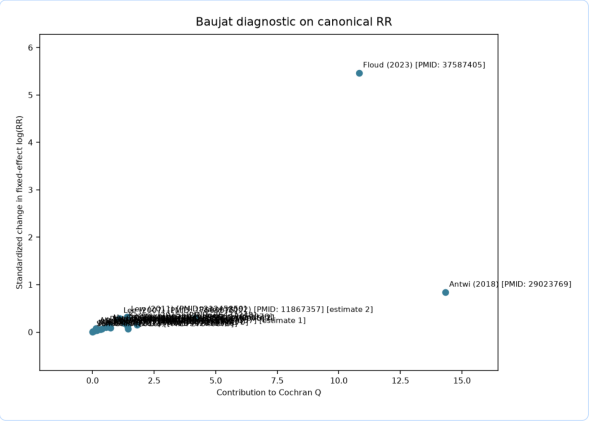
Egger's test indicates significant funnel plot asymmetry (p=0.0019) for the meta-analysis of Alcohol and Renal cell carcinoma. This finding suggests the presence of small-study effects, which may arise from publication bias or other sources of heterogeneity related to study size. The pooled estimate should therefore be interpreted with appropriate caution.

Supplemental Figure S.8: Funnel plot and Egger's test for alcohol and renal cell carcinoma, with reading guidance and an automatically generated plain-language interpretation.

Influence & Sensitivity Analysis

Baujat Plot (Influence)

Download



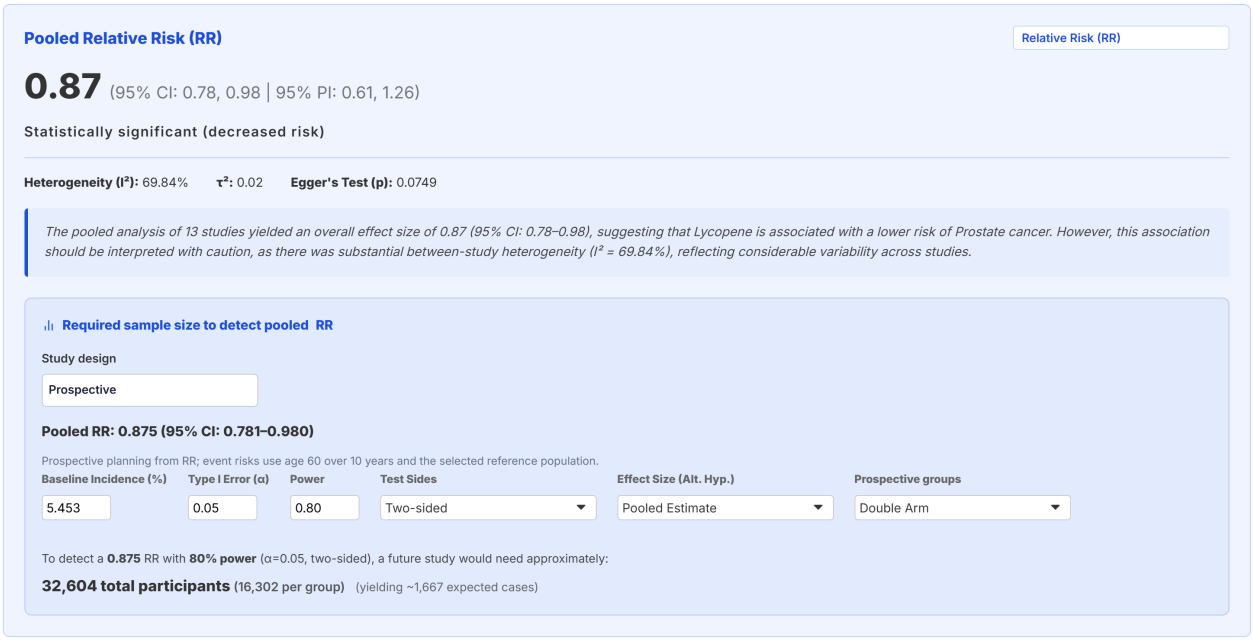
Identifies studies that contribute most to overall heterogeneity vs. individual influence on the pooled estimate.

Sensitivity (Leave-One-Out)

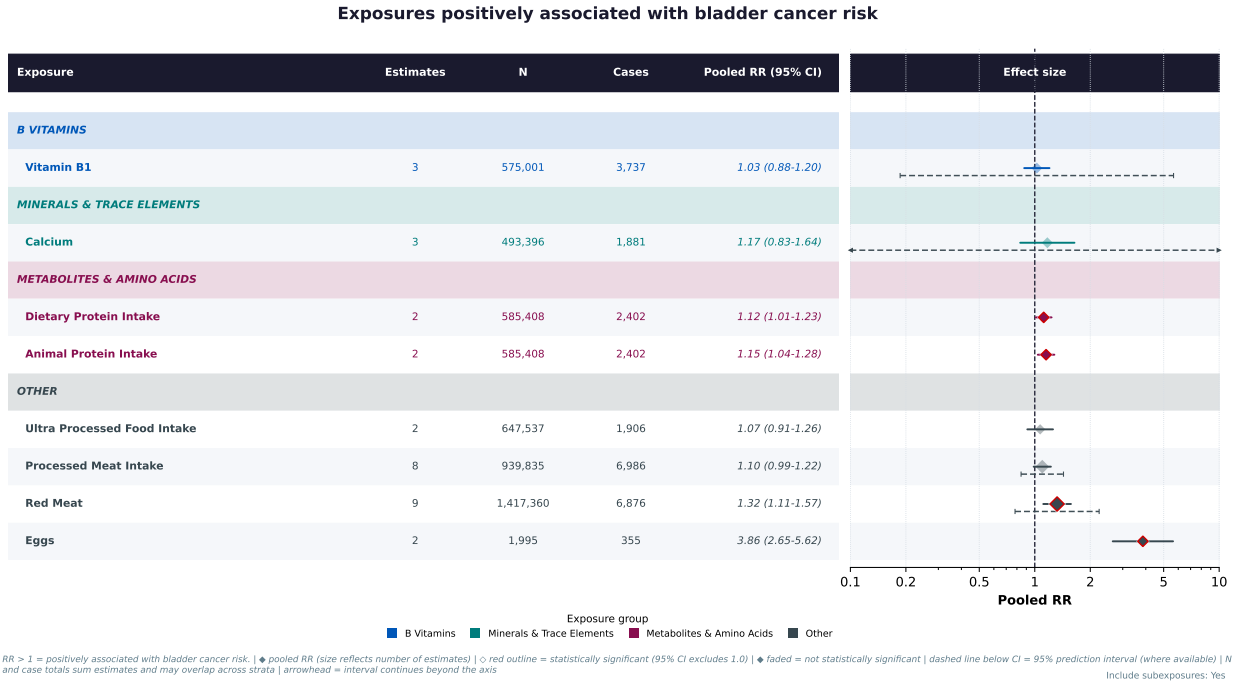
STUDY OMITTED	POOLED ES (95% CI)
Parker (2002) [PMID: 11867357]	0.72 (0.67, 0.77)
Antwi (2018) [PMID: 29023769]	0.75 (0.70, 0.79)
Goodman (1986) [PMID: 3776975]	0.72 (0.67, 0.77)
Greving (2007) [PMID: 17653076]	0.72 (0.67, 0.78)
Lee (2006) [PMID: 16775182]	0.72 (0.67, 0.77)
Karami (2015) [PMID: 25431248]	0.72 (0.67, 0.77)
Setiawan (2007) [PMID: 17656615]	0.72 (0.67, 0.77)
Macleod (2013) [PMID: 23665301]	0.72 (0.67, 0.77)
Schouten (2008) [PMID: 19064569]	0.72 (0.67, 0.77)
Hu (2008) [PMID: 18420355]	0.72 (0.67, 0.77)
Hu (2008) [PMID: 18420355]	0.72 (0.67, 0.77)
Pelucchi (2008) [PMID: 18187482]	0.72 (0.67, 0.77)
Rashidkhani (2005) [PMID: 15957170]	0.72 (0.67, 0.77)
Lew (2011) [PMID: 21245859]	0.72 (0.66, 0.77)
Lee (2007) [PMID: 17505075]	0.72 (0.66, 0.77)
Minami (2021) [PMID: 34490717]	0.72 (0.66, 0.77)
Allen (2011) [PMID: 21407222]	0.71 (0.66, 0.77)
van de Pol (2021) [PMID: 32638386]	0.71 (0.66, 0.77)
Schouten (2016) [PMID: 27340879]	0.71 (0.66, 0.77)
Floud (2023) [PMID: 37587405]	0.70 (0.65, 0.74)
Parker (2002) [PMID: 11867357]	0.70 (0.66, 0.75)

Red pooled estimates have an unrounded 95% CI excluding the null after the named study is omitted

Supplemental Figure S.9: Influence and sensitivity analyses for alcohol and renal cell carcinoma: Baujat plot and the complete list of 21 leave-one-out pooled estimates (shown in full here; the list scrolls in the application).



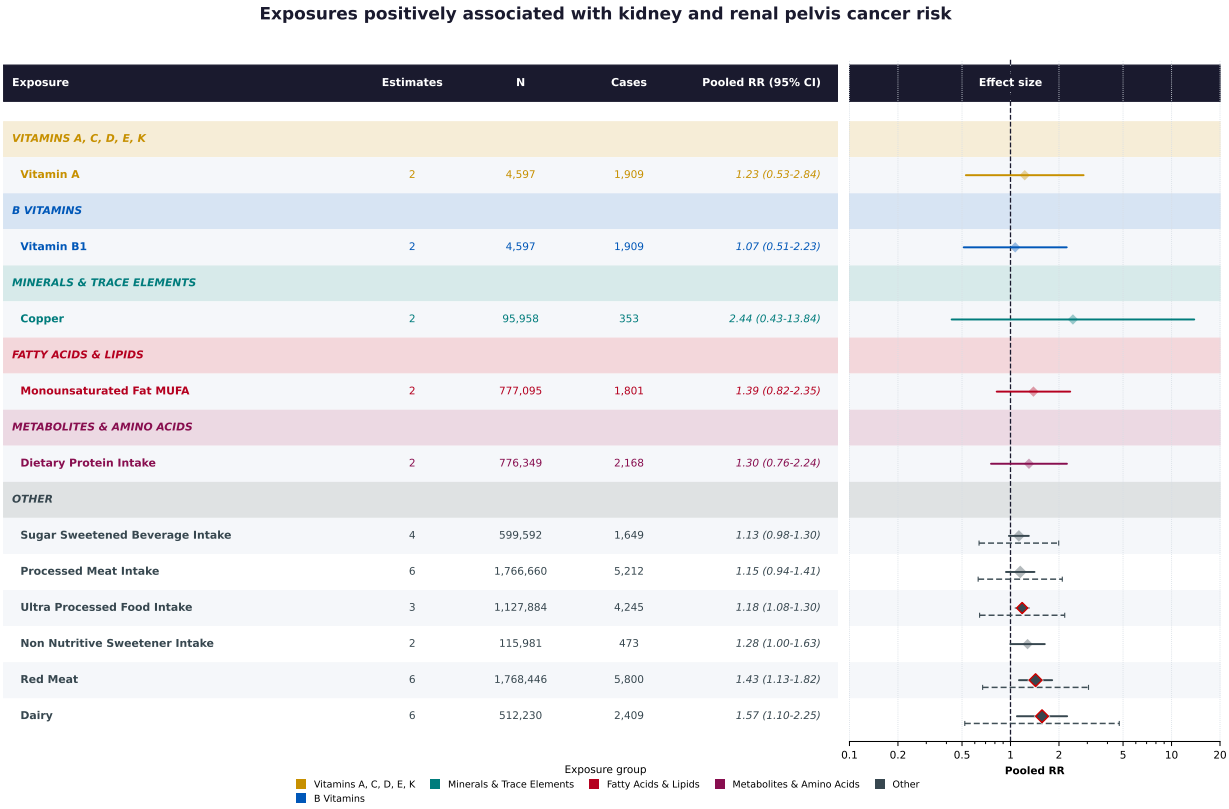
Supplemental Figure S.10: Pooled RR and sample-size planning for dietary lycopene intake and prostate cancer (Exposure Measurement filter set to dietary intake; 13 estimates), using the SEER reference risk of 5.453% for men aged 60 years followed for 10 years. The planner uses the unrounded pooled RR, displayed as 0.875.



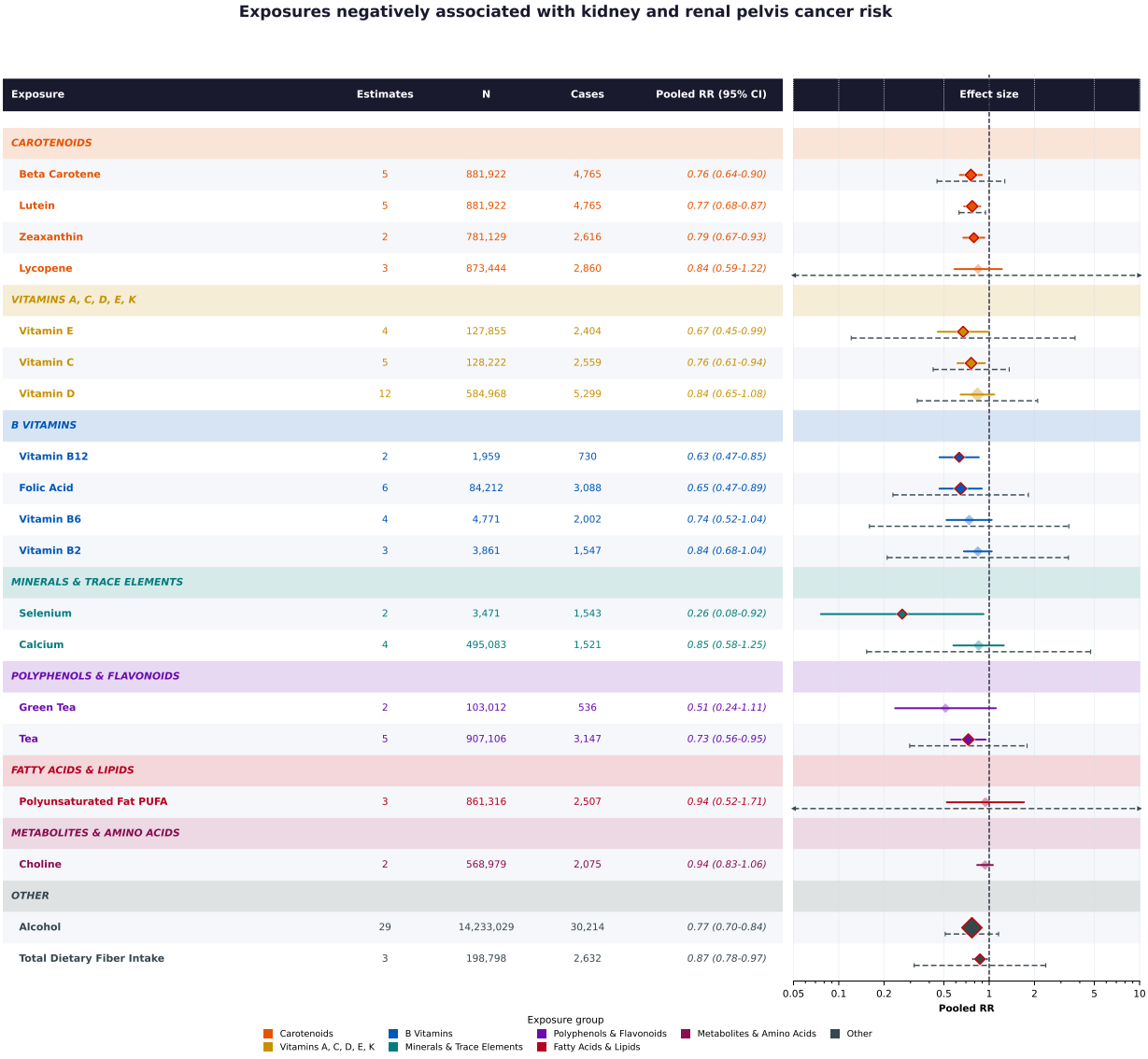
Supplemental Figure S.11: Summary forest plot of nutritional exposures with pooled risk ratios above 1.00 for bladder cancer in METAURO (at least two pooled estimates per exposure; default settings). Each row reports the number of pooled estimates (Estimates), total participants (N), and total cases (Cases). Diamonds with a red outline mark 95% confidence intervals that exclude 1, and dashed lines show 95% prediction intervals. Colors denote exposure groups.



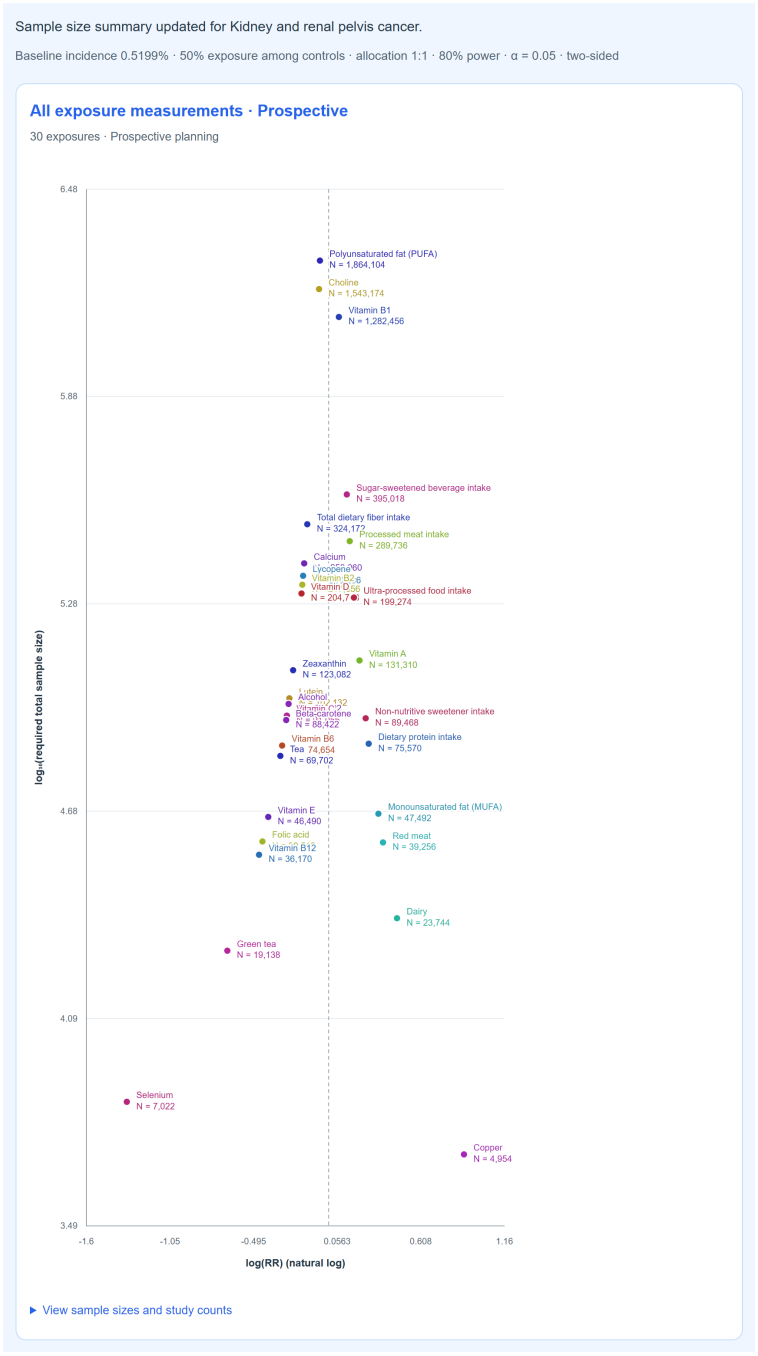
Supplemental Figure S.12: Summary forest plot of nutritional exposures with pooled risk ratios below 1.00 for bladder cancer in METAURO (at least two pooled estimates per exposure; default settings). Each row reports the number of pooled estimates (Estimates), total participants (N), and total cases (Cases). Diamonds with a red outline mark 95% confidence intervals that exclude 1, and dashed lines show 95% prediction intervals. Colors denote exposure groups.



Supplemental Figure S.13: Summary forest plot of nutritional exposures with pooled risk ratios above 1.00 for kidney and renal pelvis cancer in METAURO (at least two pooled estimates per exposure; default settings). Each row reports the number of pooled estimates (Estimates), total participants (N), and total cases (Cases). Diamonds with a red outline mark 95% confidence intervals that exclude 1, and dashed lines show 95% prediction intervals. Colors denote exposure groups.



Supplemental Figure S.14: Summary forest plot of nutritional exposures with pooled risk ratios below 1.00 for kidney and renal pelvis cancer in METAURO (at least two pooled estimates per exposure; default settings). Each row reports the number of pooled estimates (Estimates), total participants (N), and total cases (Cases). Diamonds with a red outline mark 95% confidence intervals that exclude 1, and dashed lines show 95% prediction intervals. Colors denote exposure groups.



Supplemental Figure S.15: Sample size summary plot for planning prospective studies of kidney and renal pelvis cancer. Each point represents an exposure supported by at least two studies and is labeled with the exposure name and required sample size (N). The horizontal axis shows the natural logarithm of the pooled risk ratio, and the vertical axis shows the total sample size required for a new prospective study under default settings.

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